



Faculty of Engineering,
Built Environment and
Information Technology

Multi-Channel Signal Processing for Impulsive Fault Detection in Rotating Machinery

MRN 422

Final Report

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Executive Summary

This research project aimed to develop a robust vibration-based fault detection method for rolling element bearings operating under impulsive conditions. Modern industrial machinery is often equipped with attachment points for multiple accelerometers, yet many diagnostic methods still rely on a single accelerometer. The purpose of this work was to evaluate whether using multiple sensor vibration measurements, combined with advanced signal processing and statistical modelling, can reliably improve fault detection during impulsive disturbances.

A complete detection pipeline was developed and implemented in this project. A lumped-parameter numerical bearing model was developed to simulate healthy and faulty vibration signals under controlled conditions. Multi-sensor vibration signals were generated by placing virtual accelerometers at different locations and orientations. These signals were then processed using an automatic filtering process based on the Power Envelope Spectrum and the PESOgram to isolate fault-related frequencies. A set of features based on the Power Envelope Spectrum and sparsity measures of the envelope spectrum was extracted from each filtered signal, followed by feature-level fusion using weighted-PCA. A Gaussian Mixture Model was then trained on healthy data to detect deviations from a learned healthy behaviour associated with bearing faults. The same pipeline was applied to experimentally gathered data from a multi-sensor test bench.

The results from both numerical and experimental investigations show that multi-sensor configurations provide meaningful improvements in detection performance. Sensors aligned with the applied load consistently produced clearer fault signatures, even when placed farther from the faulty bearing. This shows that orientation is more important than proximity to the fault. The detection model performed well under normal conditions, accurately separating healthy and faulty behaviour across all speeds and fault severities. This research, however, showed that the detection pipeline relies heavily on the filtering stage. When random impulses are not fully removed, the extracted feature can become corrupted, causing the GMM to misclassify healthy signals as faulty.

This research concludes that multi-sensor vibration analysis significantly improves the robustness of bearing fault detection, provided that the impulsive events are effectively suppressed. The developed method reliably performs under normal and most impulsive noise conditions and provides clear benefits over single-sensor methods. To improve the performance of the method under harsh and impulsive conditions, several recommendations are proposed: refine the PES/PESOgram automatic filtering method to improve the suppression of the random impulses, investigate additional or alternative features that remain stable when the filtering is not perfect, or investigate alternative filtering techniques that may work better under impulsive conditions.

Research Project Compliance Matrix

Requirement	Project report		Completion Status (Yes/No/In Progress)
	Section	Page	
1: Literature Review			
Review numerical bearing vibration modelling techniques.	2.2	4	Yes
Review signal processing and multi-sensor fusion	2.3-2.4	5-9	Yes
Review techniques to detect and diagnose bearing faults	2.5	11	Yes
2: Numerical Modelling and Simulation			
Develop a numerical model to simulate vibrations of both healthy and faulty bearings.	4.1-4.3	24-27	Yes
Simulate outer-race bearing faults under various conditions.	4.4	28	Yes
Simulate multi-sensor vibration signals.	4.4	28	Yes
Incorporate noisy signals.	4.4.2	29	Yes
Ensure that the simulated vibration signals are plausible.	4.4	28	Yes
3: Fault Detection Method			
Implement signal processing-based methods to refine and extract features from the vibration signal.	3.2-3.3	16-19	Yes
Implement a strategy to fuse data from multiple sensors.	3.4-3.5	21	Yes
Implement a detection method based on extracted signal features.	3.6	22	Yes
Evaluate diagnostic performance based on the simulated vibration signal.	4.7	34	Yes
4: Experimental Testing			
Collect multi-sensor vibration data from the controlled test rig with healthy and damaged bearings.	5.2	39	Yes
Introduce impulsive conditions to the testing setup.	5.3	41	Yes
Apply the developed fault detection method to the collected multi-sensor vibration signals.	5.4	42	Yes
5: Performance Comparison			
Compare single- and multi-sensor fault-detection performance under impulsive conditions.	6	48	Yes
Compare numerical predictions with experimental results.	6	48	Yes
Identify limitations of fault detection methods.	5.4-6	42-48	Yes

Acknowledgements

I would first like to thank my supervisor, Dr S. Schmidt. His guidance and support throughout this project have been invaluable, and I would not have been able to achieve this without him. I would also like to thank Mr Breghtenbach for his assistance during the experimental phase, always willing to help wherever he could.

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1 Introduction

1.1 Background

In many industries (e.g., power generation and mining), rotating machinery plays a major role in ensuring operations run smoothly. The rotating machinery is often operating continuously under harsh conditions. Equipment such as electric motors, pumps, and gearboxes [3] regularly experiences high loads, variable speeds, contamination, and large external impulsive forces. Failures in these machines often occur in components such as rolling-element bearings and gears, and can result in unplanned downtime. This results in economic losses and potential safety risks [4].

To reduce downtime and safety risks, vibration-based condition monitoring techniques are often employed to assess machine health and detect early signs of failure. These techniques rely on the fact that mechanical faults introduce characteristic vibration signatures, allowing for the early detection of critical faults [5]. Traditional condition monitoring techniques include time-domain indicators, frequency-domain analysis, and envelope analysis. These methods have proven useful under steady-state and controlled operating conditions [6].

When this machinery is used in real-world industrial environments, it often operates under harsh, impulsive conditions caused by external impacts. These impulsive conditions have the potential to hide or distort the fault-related vibration signatures, reducing the reliability of conventional condition monitoring methods [7]. This means there is a growing need for a robust diagnostic approach capable of detecting damage under harsh conditions.

The rotating machinery used in modern industry has been fitted with multiple condition-monitoring sensors, typically accelerometers mounted at different locations and orientations. Multi-sensor measurements have the potential to provide more in-depth information about the machine and improve fault detection [8]. Current fault detection techniques do not incorporate information from multi-sensor setups because it is unclear how to effectively combine the sensor data.

1.2 Problem Statement

Most modern bearing-fault detection methods rely on single-sensor vibration analysis, even though machinery can provide multiple accelerometer signals. Existing approaches offer limited guidance on combining multi-sensor information, and single-sensor performance tends to degrade under noisy and impulsive conditions. It therefore remains unclear if the robustness from a multi-sensor system is due to the number of sensors or the fusion strategy. This research aims to address this by evaluating how sensor count and fusion methods influence the robustness of a fault detection method.

1.3 Report Overview

This report outlines the development of a vibration-based bearing fault detection method designed to operate under harsh, impulsive conditions. The structure of this report follows a clear progression from theory to the development of a numerical model.

Section 2 provides a comprehensive literature review covering key aspects of multi-sensor vibration fault detection. The review establishes a strong theoretical foundation for developing a statistical cyclostationary detection method and guides the selection of appropriate feature-level data fusion techniques.

Section 3 describes the proposed fault detection pipeline and methodology. It outlines the full signal processing methods, including automatic signal processing using PES and PESOgram, feature extraction and fusion with weighted PCA, and the use of a Gaussian Mixture Model for anomaly detection.

Section 4 presents a bearing vibration model that is based on a lumped-parameter dynamic model. It includes modelling of bearing kinematics and Hertzian contact forces, and the generation of both healthy and faulty vibration signals. Single- and multi-accelerometer configurations, including noise, are simulated to evaluate the detection pipeline under controlled conditions.

Section 5 presents the experimental investigation conducted on a multi-sensor bearing test bench. The experimental setup, test cases, and data-collection procedure are described, followed by the application of the full detection pipeline to real vibration measurements.

2 Literature Review

Rolling-element bearing faults are among the main causes of rotating machinery failure, and vibration-based fault detection is an effective means of detecting them. This literature review summarises the models available to simulate bearing vibrations. Surveys advanced signal processing methods for extracting fault-related features. Evaluates multi-sensor fusion strategies for robust detection, as well as the methods for detection.

2.1 Rotating Machine Faults and Vibration-Based Condition Monitoring

Rolling Element Bearings are critical components in most rotating machinery and are among the most common causes of failure. Vibration-based condition monitoring is commonly used for bearing fault detection as it is sensitive to impulsive responses generated by defects. Bearing faults typically include damage to the inner race, outer race, or rolling elements [9].

Localised bearing defects generate periodic impulse forces as the rolling elements in the bearing pass over the damaged area. The impacts introduce small shock excitations that cause a high-frequency resonance of the bearing housing and the mechanical path between the bearing and the vibration transducer. The typical signals picked up from the vibration transducer for common rolling element bearing faults are seen in Figure 1. This vibration behaviour enables the distinction of bearing faults from other machine faults, such as imbalance and misalignment. [9]

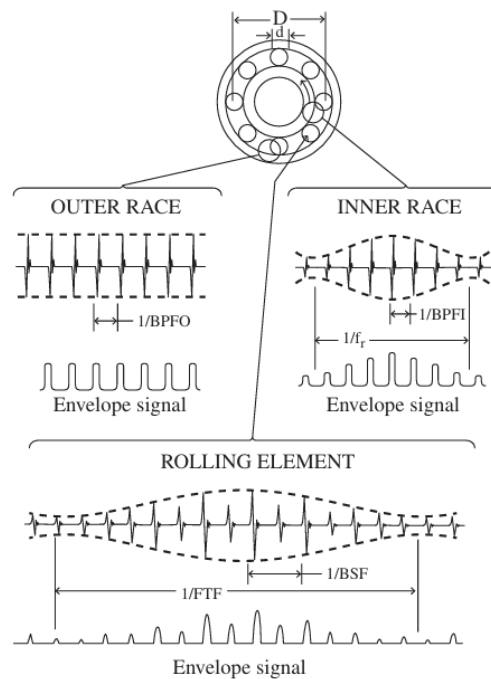


Figure 1: Typical signals and envelope signals from faults in rolling element bearings [9]

The frequency of the impulsive excitation from the bearing fault is governed by its velocity and rotational speed. This means that a fault will produce characteristic frequencies that allow the defect location within the bearing housing to be pinpointed. These characteristics include the ball pass frequency of the outer race (BPFO), the ball pass frequency of the inner race (BPFI), the ball spin frequency (BSF), and the fundamental train frequency (FTF). These characteristic frequencies form the basis of theoretical models for vibration-based rolling-element bearing fault detection. They also help to identify the location and nature of the defect [9].

2.2 Numerical Bearing Vibration Model

To create a robust and reliable fault-detection method, it is important to test and refine it with vibration data. However, getting experimental vibration data that includes all the expected defects is often expensive and time-consuming. This means that a numerical simulation of the vibration signal is useful for ensuring the method's robustness and for training machine learning models.

2.2.1 Phenomenological Models

Phenomenological bearing-fault vibration models describe the vibration response with the use of observable signal characteristics rather than the mechanical behaviour of the bearing [10]. These models represent the defect as a sequence of periodic or quasi-periodic impulsive events. The amplitudes and spacing of the signal vary to account for slip and fluctuations in conditions [11]. As the model is based directly in the signal domain, it reflects the cyclostationary behaviour typically observed in bearing-fault vibration signals.

Phenomenological models are often used because they allow precise control over the signal features that influence fault detection performance. Researchers can independently adjust impulse timing, damping, frequency, and noise intensity, for example. They are computationally efficient and allow direct manipulation of signal characteristics, making them a good option for algorithm benchmarking.

However, phenomenological models lack physical grounding, as they do not represent rolling-element contact mechanics or physical transmission paths. This means that it is difficult to accurately reproduce the propagated vibration signal at different signal locations. Limiting its usefulness in a multi-sensor study where the vibration responses at each sensor rely on its relative position to the fault.

2.2.2 Lumped-Parameter Models (LPM)

Lumped-Parameter Models (LPM) provide a mechanical representation of a rotor-bearing-housing system [12]. In these models, the system is represented by masses, springs, dampers and rolling elements that approximate Hertzian behaviours [13]. The simplified spring-mass-damper system that represents the bearing model is shown in Figure 2. LPM models simulate how defects physically alter the contact forces between the raceways and rolling elements, and monitor how these forces excite the structure of that machine [12].

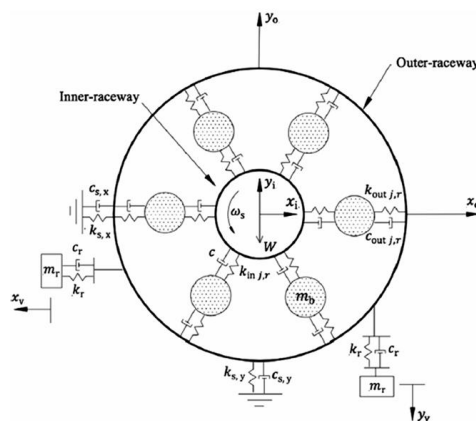


Figure 2: Lumped-Parameter Model bearing representation [13]

Because LPMs incorporate realistic kinematics, geometries, and contact mechanics, they naturally account for changes in transmission paths, sensor placement, and structural dynamics in the simulated signals [12]. The physical grounding makes it useful for multi-sensor studies, as the sensors' relative positions influence the measured signals. The main disadvantages of LPMs are their greater complexity and computational costs.

2.3 Advanced Signal Processing Methods

Bearing fault vibration signals from rotating machinery contain periodic components, resonant responses, and transient events. To extract information from these signals, various signal processing methods are available. These methods can be separated into Time Domain, Frequency Domain, Time-Frequency Domain, and Time-Scale Domain methods [14]. These methods are shown in Figure 3. Each method allows the extraction of different features for fault detection.

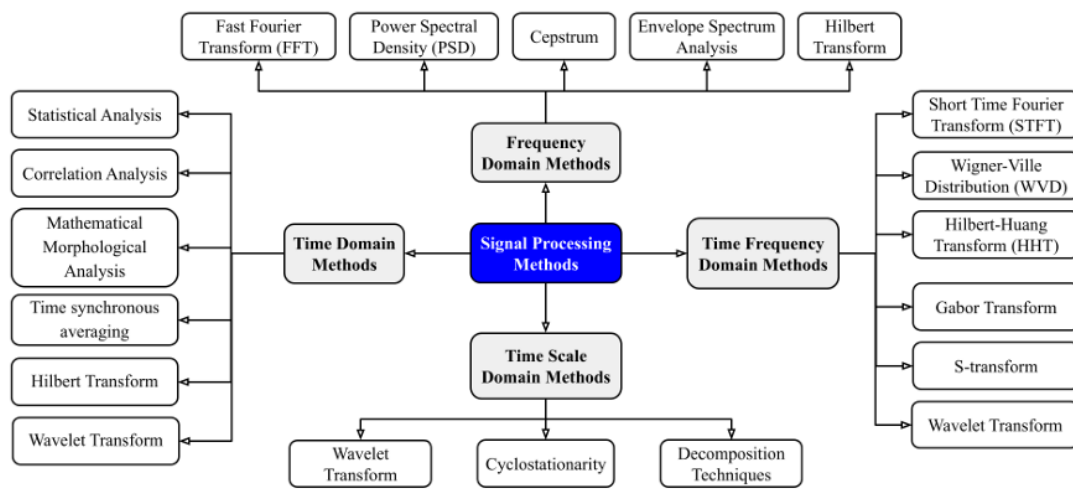


Figure 3: Signal Processing Methods [14]

2.3.1 Time Domain Analysis

Time domain methods analyse the raw vibration signal without transforming it into the frequency or time-frequency domains [14]. These methods aim to quantify the statistical behaviour of the signal over time. They capture characteristics such as energy, impulsiveness, variability, and amplitude distribution [15]. Because bearing faults often alter the amplitude and shape of the vibration signal, time-domain indicators provide a simple and meaningful set of features useful for fault detection.

These methods are widely used in bearing fault diagnostics due to their simplicity and effectiveness. Although time-domain features do not include any frequency-specific information, they still provide vital information for fault detection. Common time domain features used in bearing fault diagnostics are the Clearance Factor, Skewness, and Kurtosis [15].

2.3.2 Frequency Domain Analysis

Frequency-domain methods analyse vibration signals by examining their frequency components rather than their raw time-domain waveforms. Instead of analysing how the signal evolves over time, these methods analyse how fast the frequency changes [14]. This is done by observing dominant frequencies, harmonics, and spectral patterns [14]. This is useful for bearing vibration

fault diagnostics, as faults generate characteristic frequencies. Transforming the signal to the frequency domain makes it easier to detect periodic components, sidebands, and resonance peaks that are difficult to detect in the time domain [16].

Frequency-domain analysis is widely used in bearing fault vibration monitoring because it is easily interpreted and provides valuable fault-related features. Methods such as the Fast Fourier Transform (FFT) and Power Spectral Density (PSD) are often used when identifying fault-related frequency components [16]. Frequency-domain methods cannot capture transient or nonstationary behaviour as well as other methods; however, they still provide useful information. Envelope spectrum analysis is also commonly used, as it enhances weak modulated fault signatures that may be hidden in broadband vibration noise. It is obtained by computing the FFT of the signal's envelope. As described by Peeters et al. [17] it is a powerful tool for sparsity-based features.

2.3.3 Time-Frequency Analysis

Time-frequency methods analyse the vibrations by studying how their frequency content changes over time. Instead of assuming that the vibration signal is stationary, these methods provide a time-frequency representation to capture transient events, impulses and spectral patterns [14]. Representing the signal in both the time and frequency domains facilitates the detection of nonstationary and speed-dependent frequency shifts. This is useful for bearing vibration fault diagnostics, as the faults often produce short-duration impacts that are not visible in the time or frequency domains.

Time-frequency analysis is widely used in bearing fault detection systems as it provides a detailed view of the signal's behaviour under varying operating conditions. Methods such as the Short-Time Fourier Transform (STFT) and wavelet transform are often applied to identify transient fault signatures and track how frequency components evolve [18]. These methods offer improved insights compared with time- or frequency-domain analysis. However, they do include trade-offs in resolution and can be sensitive to parameter selection [18]. Other approaches, such as the S-transform, aim to improve signal concentration [19]. However, STFT and wavelet transforms are still the most common time-frequency methods used [18].

2.3.4 Time Scale Analysis

Time-scale analysis examines the vibration signal by decomposing it across multiple characteristic scales. This enables the extraction of multiple features that vary with amplitude or duration [14]. Unlike time-frequency methods that focus on how the spectral content varies over time, time-scale methods examine how the signal behaves at different resolutions. This makes them effective for impulsive or multi-scale patterns, as the frequency resolution is higher when needed [14].

In vibration-based bearing fault detection, time-scale analysis is widely used as it provides a flexible framework for detecting fault signatures. Wavelet-based scale analysis [18] and empirical mode decomposition (EMD) [20] methods are often applied to separate modes and show fault-related impulses. EMD methods achieve this with the use of Intrinsic Mode Functions (IMFs) that break a complex signal into simpler components [20]. These techniques are well-suited to nonstationary signals, as they can reveal subtle features that might be missed with time-, frequency-, or time-frequency-domain methods. However, they introduce challenges such as mode mixing and parameter selection. Despite this timescale, methods provide a robust tool for extracting fault-related features for bearing fault detection systems.

2.3.5 Cyclostationary Analysis

Cyclostationary analysis examines vibration signals in which their statistical properties vary periodically over time [21]. Making it effective for detecting modulated or periodically excited fault signatures. Unlike the time- or frequency-domain methods that assume the signal to be stationary, cyclostationary methods use the fact that bearing faults create characteristic defect frequencies that produce periodic modulations in the vibration signal [22]. Analysing the second-order periodicity through cyclic autocorrelation and cyclic spectral density, cyclostationary analysis methods can isolate the modulating components [21].

Cyclostationary is used in bearing fault diagnostics as it provides a strong method that allows for the identification of amplitude- and frequency-modulated signals associated with bearing fault vibrations [21]. The cyclic spectral maps reveal fault-related harmonics and sidebands that may not be visible in ordinary spectra. Cyclostationary analysis requires careful consideration of cyclic frequencies and averaging parameters to provide trustworthy results. If the input parameters are correctly chosen, cyclostationary analysis provides a robust method for extracting weak, periodic fault signatures [23].

2.3.6 Overview of Analysis Methods

Table 1 provides an overview of commonly used signal processing methods for vibration-based condition monitoring and bearing fault detection. Each signal processing method is followed by its advantages, limitations, and useful features. The comparison is important for selecting the best methods to detect weak or transient fault signatures in real-world vibration signals.

Table 1: Comparison of Advantages, Limitations, and Features Across Signal Processing Methods

Signal Processing Method	Advantages	Limitations	Useful Features	Ref.
Time Domain Analysis	Fast and easily interpretable	Only simple features	Clearance Factor, Skewness, Kurtosis, and RMS	[14], [15]
FFT	Clear frequency peaks and well-established	Poor for transients, sensitive to noise, and spectral leakage	Peak amplitudes, harmonic levels, sideband spacing, and band energy	[14], [16]
Envelop Spectrum	Reveals weak bearing fault signatures	Highly sensitive to filter selection	$\frac{l_2}{l_1}$ norm, spectral negentropy and hoyer index	[17]
PSD	Noise-robust, smooth spectral estimates, highlight resonance bands and are good for energy analysis	Assumes stationary, averaging smears weak faults, lower resolution	Band power, resonance energy, spectral slope, and PSD entropy	[14], [16]
STFT	Time-frequency view, simple, and intuitive	Fixed window, poor transient resolution, parameter-sensitive	Spectrogram energy, ridge peaks, time-varying frequency content, and kurtogram	[14], [18], [19], [24]
Wavelet Transform	Good for transients, multi-resolution, and flexible	Wavelet choice sensitive	Wavelet energy, wavelet entropy, and scale-dependent kurtosis	[14], [18]
S-Transform	Time-frequency localisation, frequency-dependent window, and improved concentration	Can blur sharp transients	S-transform magnitude, time-varying spectral peaks, and localised energy features	[14], [18], [19]
EMD	Adaptive decomposition and good for transients	Mode mixing and sensitive to noise	IMF energy, IMF kurtosis and IMF entropy	[14], [20]

Cyclostationary Analysis	Detects periodic modulations and is robust to noise	Careful cyclic-frequency selection and interpretation can be hard	Cyclic spectral density and cyclic energy	[14], [21], [22], [23]
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2.3.7 Signal Processing for Impulsive Conditions

Unlike bearing-fault vibration impulses that repeat at characteristic frequencies, random impulses have no periodic structure and often include heavy-tailed, non-Gaussian behaviour [25]. The presence of random impulses can distort vibration analysis, since many signal-processing methods rely on measures of impulsiveness, such as kurtosis. This can result in the envelope spectrum being contaminated and the weaker periodic fault-related impulses being masked.

To minimise the effect of random impulses on fault detection, Yang et al. [17] developed a preprocessing technique to identify and remove random, non-cyclic impulsive noise from the signal. Their method divides the signal into short segments and uses kurtosis to identify those showing abnormal impulsive behaviour. For each section showing abnormal behaviour, they check how the impulse lifts the entire envelope spectrum and construct an impulse indicator from the squared envelope spectrum. By sliding the window across the signal, they can detect the start and end of a random impulse by looking for an upward trend within the window. After the start and end of the impulse are identified, they remove the impulse and replace it with Gaussian noise that matches the surrounding signal. This produces a cleaned signal that retains the signal's realistic statistical properties. Their results show that this approach accurately and efficiently cancelled out the non-cyclic impulsive noise and improved the visibility for the true cyclic fault-related impulses. The effects of this are shown in Figure 4.

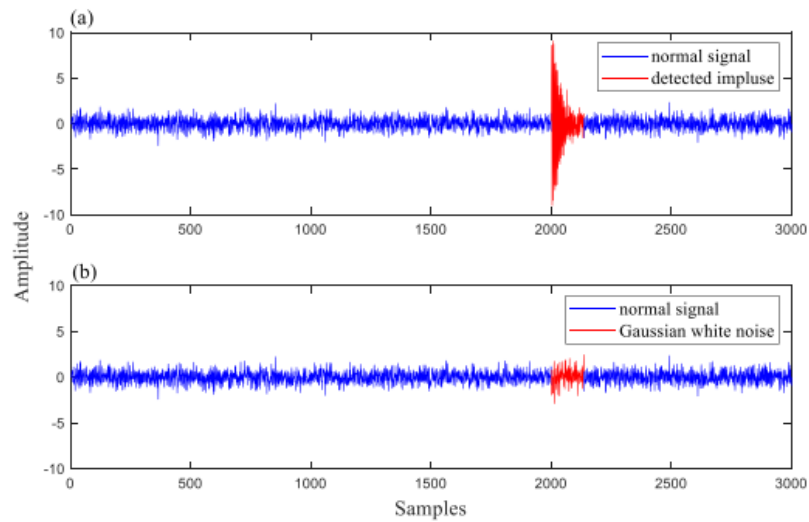


Figure 4: a) Signal Including Impulse, b) Signal after Impulse Cancellation [26]

Another method developed by Chen et al. [26] follows a fundamentally different approach to minimise the effect of random non-cyclic impulsive noise on the signal. Rather than removing the random impulse from the signal, they propose an enhanced envelope analysis method to improve the detection of cyclostationary fault signatures in rolling bearings, particularly in the presence of random impulsive noise. They start by generalising envelope signals using a simplified Box-Cox transformation, producing a family of generalised envelope spectra (GESs) that are proposed to reveal cyclostationarity. GESs with different transformation parameters offer

distinct advantages for random impulse noise and Gaussian background noise, enabling the method to adapt to varying conditions. To increase robustness, they introduce the Product Envelope Spectrum (PES), which multiplies multiple GESs to combine their distinct advantages and highlight periodic fault-related components. Finally, they developed the PESOgram, which is an optimisation-based enhancement that improves the accuracy and stability of PES. When testing the PESOgram with simulated and experimental railway bearing data, they find that it improves the visibility of fault-related characteristic frequencies and often outperforms state-of-the-art envelope analysis methods in noisy environments.

2.4 Multi-Sensor Condition Monitoring

Accelerometers mounted on the rotating machine housing record vibrations that include information on bearing conditions. Many modern systems use a single accelerometer for simplicity; however, this approach is susceptible to sensor misplacement, which can cause important fault measurements to be missed. A multi-sensor layout can capture signals from different structural paths and orientations, improving fault detection. The objective is to minimise the number of sensors, so the information from additional sensors should be as effective as possible. This section will cover the different fusion levels and which fusion strategy will be most effective.

2.4.1 Multi-Sensor Fusion Levels

While using multiple accelerometers yields a more well-defined vibration signal, the effectiveness of a multi-sensor fault detection system depends heavily on the chosen signal fusion method. Multi-sensor fusion aims to combine individual sensor signals to improve the accuracy, reliability, and robustness of the fault detection system. There are three main types of data fusion methods commonly used in vibration-based fault detection: data-level, feature-level, and decision-level fusion[27].

2.4.1.1 Data-Level Fusion

Data-level fusion combines raw data from multiple sensors before feature extraction. Data-level fusion is the lowest level of data fusion, in which homogeneous data from multiple sensors are combined to create a single data source for feature extraction [27]. Since the raw data is combined before any signal processing, no information is lost during fusion. Older methods for data-level fusion include weighted averaging and Kalman filtering; more recent work utilises multi-channel Convolutional Neural Networks (CNNs) that can learn directly from the raw fused vibration signal. The limitations include high computational costs, redundancy in raw data that reduces accuracy, and a restriction to only one sensor type.

2.4.1.2 Feature-Level Fusion

Feature-level fusion extracts features from each sensor independently and then combines them. This creates a higher-level representation of the signal by combining features from multiple sensors[27]. This is the level commonly used because it naturally aligns with signal-processing diagnostic methods. Common methods for feature-level fusion include concatenation, canonical correlation analysis, and deep learning-based fusion. This level offers a fair balance between detailed information and computational efficiency; however, care is required when managing high-dimensional fused feature sets.

2.4.1.3 Decision-Level Fusion

Decision-Level fusion combines the outputs from multiple sensor-specific diagnoses. This level will combine decisions from multiple sensors to make a final decision [27]. Common decision-

level fusion methods include majority voting, weighted voting, Dempster-Shafer theory, Bayesian fusion, and fuzzy logic. This level excels in noisy, uncertain environments and is ideal for safety-critical applications. The main drawback is a loss of detail in the signal, meaning subtle fault signatures are often missed.

2.4.2 Multi-Sensor Fusion Strategies

This section reviews multi-sensor fusion strategies for bearing fault detection. Table 2 summarises the commonly used fusion and dimensionality-reduction methods in the literature. For each method, Table 2 outlines its core principle, advantages and disadvantages, and the level of fusion at which it is typically applied. This provides a structured method for selecting the appropriate strategy to fuse information from multiple accelerometers.

Table 2: Summary of Fusion and Dimensionality Reduction Techniques

Fusion Method	Core Idea	Advantages	Disadvantage	Fusion Level	Ref.
Principal Component Analysis (PCA)	Linear orthogonal projection that captures maximum variance	Fast, non-iterative, and reduces redundancy and noise	Linear only and sensitive to feature scaling	Feature	[28], [29], [30], [31]
Linear Discriminant Analysis (LDA)	Supervised linear projection that maximises class separability	Produces projections that maximise separability for labelled faults and is effective for classification	Suffers from singularity and assumes class scatter is appropriate	Feature	[28], [29], [31]
Independent Component Analysis (ICA)	Decomposes mixed observations into statistically independent components	Can separate overlapping signals and filter noise	Assumes independent non-Gaussian sources	Feature/ Data	[28], [32]
Canonical Correlation Analysis (CCA)	Finds projections of multiple sensor views that maximise mutual correlation	Explicitly fuses correlated information across features	Linear assumption, requires paired multi-view samples and sufficient data	Feature	[28], [33]
Singular Value Decomposition (SVD)	Low-rank matrix approximation that compresses the feature matrix into dominant modes	Robust linear compression and handles sparse matrices	Linear only, and the choice of retained rank affects performance	Feature	[31]
Kernel PCA (KPCA)	Apply PCA in a kernel space to capture nonlinear separability	Captures nonlinear structure and is useful when PCA fails	Requires kernel choice and parameter tuning	Feature	[28], [31]
Manifold/local nonlinear methods (LLE, ISOMAP, LTSA, LE, t-SNE)	Preserve local/ global nonlinear geometry to embed data into low-D manifolds	Captures nonlinear structure in vibration data.	Sensitive to hyperparameters, t-SNE is mainly for visualisation, ISOMAP can suffer from topology instability	Feature	[28], [31]

It is important to note that the studies done by Buchaiah and Shakaya [28] and Anowar et al [31] both use PCA as a baseline method. It is used either as a reference point for evaluating other methods or as a standard linear benchmark because of its simplicity, speed, and widespread use in feature-level fusion.

2.5 Probabilistic Detection Models

2.5.1 Multivariate Statistical Process Control

Multivariate Statistical Process Control (MSPC) builds a healthy reference database from historical vibration features and monitors new observations against that reference [34]. This is a two-stage procedure consisting of offline modelling using normal data and online monitoring of the test signal. This method uses dimensionality reduction techniques such as PCA, ICA, and HCA to represent correlated features as low-dimensional latent variables, which is useful when faulty examples are scarce [34].

To make a decision based on the low-dimensional latent variables from the MSPC model, decision metrics are used to draw a conclusion. Jin et al. [34] compare the following decision metrics:

- **Hotelling's T^2 :** This is used to quantify how far a fused feature vector projects along the directions obtained from PCA/ICA/HCA. Jin et al [34] use T^2 as the main indicator for changes that remain inside the learned healthy subspace, and use it to detect shifts in correlated behaviour. T^2 is interpreted relative to the control limits that have been derived from the healthy training set [34].
- **Squared Prediction Error (SPE)/ Q-statistic:** This measures the reconstruction error after projection into the latent subspace. SPE control limits are estimated from the healthy data and sometimes require empirical or kernel-density methods when analytical distributions are not available [34].
- **Mahalanobis-Distance (MD):** Provides a single-index alternative. However, it is less reliable when features are strongly correlated.

Jin et al [34] made the decision to create a two-complementary statistic computed on a model that is trained with healthy data. Hotelling's T^2 is used to measure the projection onto the dominant components, and the SPE/Q-statistic measures the reconstruction error. A sample is flagged as containing a fault if either control limit is exceeded.

2.5.2 Gaussian Mixture Models

Wang et al [35] propose a Gaussian Mixture Model (GMM) based anomaly detection method called Deep Autoencoder-Transformer GMM (DTGMM). In this proposed method, the GMM is used solely as a probabilistic density estimator trained on normal operating data. The purpose of the GMM is to learn the distribution of the healthy data and detect deviations from it. Because the model is trained only with normal healthy data, it performs unsupervised anomaly detection rather than fault classification.

The GMM is trained using a fused latent vector that combines a range of features into a single vector representation of normal system behaviour. This vector is used as input to the GMM, which estimates the mixture weights, means, and covariances for each Gaussian Component. The GMM can represent several healthy operating conditions and the complex variability in the given data. Making it an effective density estimator for identifying deviations from normality in the training data.

The decision metrics used:

- Negative Log-Likelihood (NLL): Wang et al [35] use the NLL from the GMM as their only anomaly score. A high NLL indicates that a tested sample lies in a low-density region of the learned distribution and is therefore abnormal.
- Kernel-density-thresholding: The thresholds for the NLL are derived from the distribution of healthy training samples with the use of kernel density estimation.

Although Wang et al [27] applied DTGMM to power plant equipment, the overall principle is independent of the input features. The GMM just models the distribution of whatever feature set it is provided with. This means it is possible to apply this anomaly-detection logic to vibration-based monitoring or to any other engineering system. As the model is trained only on healthy data, it can raise an alarm only when a fault is present, since it recognises deviations from the signal's normal patterns. Unfortunately, it cannot determine a specific defect, so additional signal processing is required.

2.6 Conclusion

The reviewed literature shows key modelling, signal processing, fusion, and detection methods that support the development of a reliable multi-sensor bearing fault detection method. The following conclusion shows how these methods align with the project and how the selected approaches are relevant.

2.6.1 Numerical Bearing Vibration Model

The modelling approach selected for the bearing vibration model is the LPM, as it aligns with the physical vibration transmission paths in rotating machinery. Because this model relies on actual contact mechanics and structural dynamics, it can simulate sensor-dependent vibration signals that are difficult to replicate with the phenomenological model. This makes it possible to see how fault-related signals propagate through different transmission paths associated with sensor orientation and placement. As this research aims to assess the usefulness of multiple sensors, the positional information available from the LPM is vital. By using an LPM, the modelling approach will more closely represent the machine's behaviour, enabling testing of fault detection methods with a realistic vibration signal.

2.6.2 Signal Processing and Feature Extraction

The selected approach will be to apply the PES and PESOgram filtering method from Chen et al. [26] to filter the signal and highlight the cyclic impulses at the expected fault frequencies. The selected feature for building the fault detection model complements the chosen filtering technique. A small set of four features is selected from either the PES or the product envelope spectrum. The chosen features are the FDSNR and the magnitude of the first harmonic from the PES; the other two features are sparsity measures, namely the $\frac{l_2}{l_1} norm$ and the spectral negentropy [36] as these provide valuable fault detection measures from the envelope spectrum. These features have been selected as they all provide meaningful physical information from the vibration signal and align well with the filtering method. A small set of features has been chosen as each provides useful information without adding unnecessary dimensionality that will later have to be managed. These few features aim to provide a low-dimensional feature set, with each providing useful information to the detection process.

2.6.3 Signal Fusion

For feature fusion, PCA is used to combine the extracted features from multiple accelerometers into a single low-dimensional feature vector. The feature weighting process described by Chaleshtori and Aghaie [37] will be applied so that features operating on different numerical ranges and with larger operating magnitudes do not dominate the PCA fit. This weighting process will ensure that all features contribute equally to the fused representation of the feature vector. However, the feature removal process they use will not be applied, as all proposed features are intended to provide meaningful information.

2.6.4 Fault detection

For the fault detection, a GMM trained on healthy vibration data will be used to detect anomalies in sample vibration signals. This follows a similar process to that used by Chaleshtori and Aghaie [37] although their model is trained on both healthy and faulty data. This does make it possible to use their model as a classification system; however, it is limited by the scarcity of vibration signals containing specific fault signatures. The GMM-based method for anomaly detection proposed by Wang et al [35] provides a robust way to model normal expected behaviour across multiple operating conditions. This makes it well-suited for any applications in which the equipment will operate under varying conditions. This aligns closely with the goals of this project, which evaluates the effectiveness of multiple vibration sensors under different operating conditions. For this reason, the selected approach will train the model only on healthy features, allowing it to function as an anomaly-detection model rather than a fault-classification model.

3 Proposed Fault Detection

This section outlines the proposed fault detection process that will later be evaluated using both numerical and experimental data. It begins by presenting the overall pipeline, from the raw input signals to a metric for detecting anomalies. The following subsections describe the stages of the process: automatic signal filtering, feature extraction, construction of feature vectors and matrices, feature weighting, and, finally, training and testing of the detection model.

3.1 Fault Detection Pipeline

This section outlines the proposed methodology for determining whether the captured vibration signal indicates a fault. Figure 5 shows the pipeline for the signal-processing and anomaly-detection procedure. The process starts with filtering each input vibration signal. For each signal obtained from the selected accelerometer combination, an automatic filtering stage is applied.

Using 1/3 binary decomposition, each signal is divided into narrower frequency bands. A band-pass filter is then applied to each of these bands. The filtered output for each band is used to calculate a Power Envelope Spectrum (PES) using a Box-Cox transformation. The Frequency Domain Signal-to-Noise Ratio (FDSNR) is found for each band, and a Product Envelope Spectral Optimisation-gram (PESOgram) is used to identify the band with the highest FDSNR. This will be the band that is selected to filter the signal.

After the automatic filtering stage, each raw vibration signal is individually filtered based on its selected frequency band. The filtered signals are then passed to the feature extraction stage. In this step, a preselected set of vibration-based scalar features is extracted from each signal. At each time step, features will be extracted from all signals and combined into a single feature vector. This process is repeated at each time step, and the feature vectors are then all added to a signal feature matrix.

At this point, the pipeline separates into two paths. The first path is used for training the fault detection model. This path is followed only when the feature matrix from the feature extraction stage contains only healthy vibration features. Limiting the model to training only on healthy features ensures that it learns normal operating conditions and can then identify anomalies in a vibration signal or combination of signals. The second path is used for any test feature matrix. This path is followed when testing the model on unseen data or when it is unknown whether the signal contains a fault. The test data may include both healthy and faulty vibration features.

The next step is to ensure that all of the extracted features are comparable. Each feature is normalised so that features with larger magnitudes do not dominate the detection process. The normalisation is achieved by computing a weighting matrix from the training data. This weighting matrix is then applied to both the training and feature matrices. This means that the training and test features are weighted consistently and can be compared.

Even though the weighted feature matrices are now comparable, they still contain all the extracted features and will therefore be high-dimensional. To reduce the dimensionality, Principal Component Analysis (PCA) is fitted to the weighted training matrix. The fitted PCA model is then applied to both the training and test matrices. After this step, the training and test matrices are directly comparable and have reduced dimensionality. This is more important when the vibration signals from multiple accelerometers are being analysed to detect a fault.

The Weighted PCA (WPCA) training matrix is used to fit a Gaussian Mixture Model (GMM). To ensure that the GMM properly represents the training data, it is important to select an appropriate number of mixture components. This is done by fitting several candidate GMMs that have different numbers of mixture components. For each candidate GMM, the Bayesian Information Criterion (BIC) is calculated. However, to prevent selecting too many mixture components and to avoid overfitting, the chosen GMM is the one where the BIC score has stabilised near its minimum.

Finally, the selected GMM is used to determine if the signal shows any signs of an anomaly. When the test data is passed to the trained GMM, it is possible to compute a Negative Log-Likelihood (NLL) score for each data point. A data point is classified as an anomaly if its NLL exceeds the threshold defined by the learned healthy training data. Although it is not possible to classify a specific fault type from the NLL score, it can detect deviations of the signal from the learned healthy behaviour.

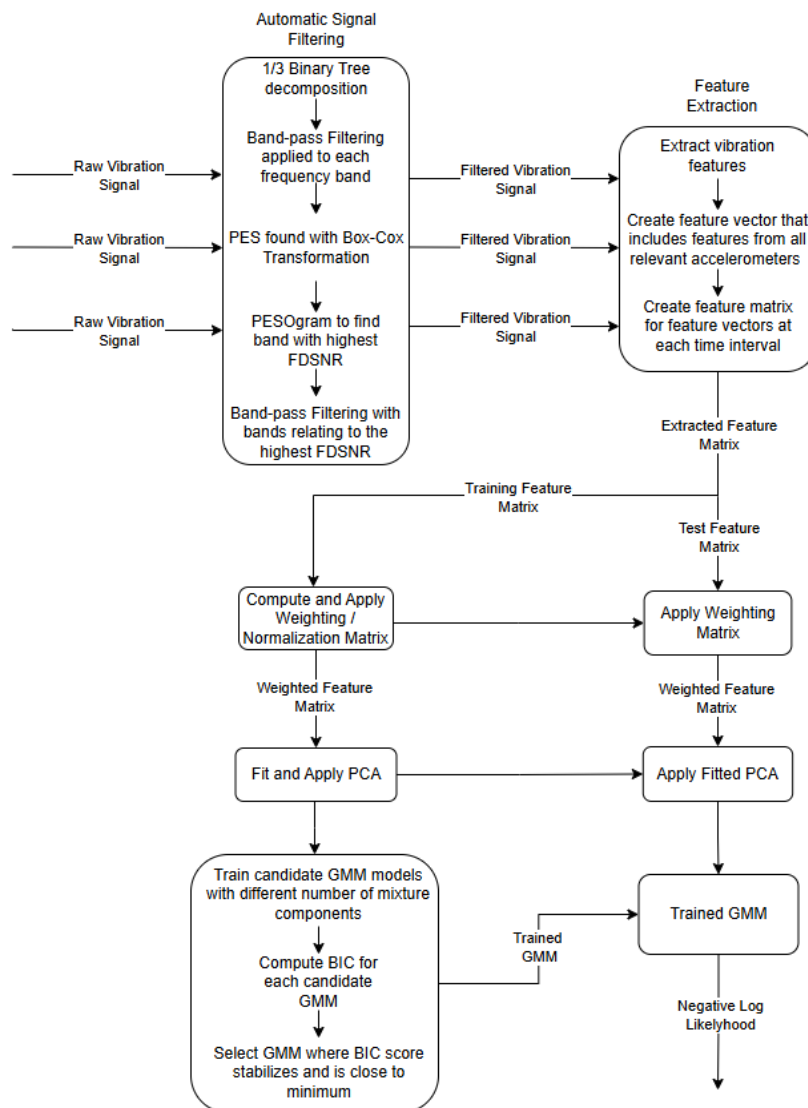


Figure 5: Proposed signal processing and anomaly detection pipeline

Figure 6 shows a vibration signal generated by the numerical bearing vibration model described in Section 4. In this section, the signal will serve purely as a visual aid to explain the filtering pipeline. Unfortunately, due to the large number of data points that are required to train the GMM,

the visual representations of the full method will be shown where relevant in the following section.

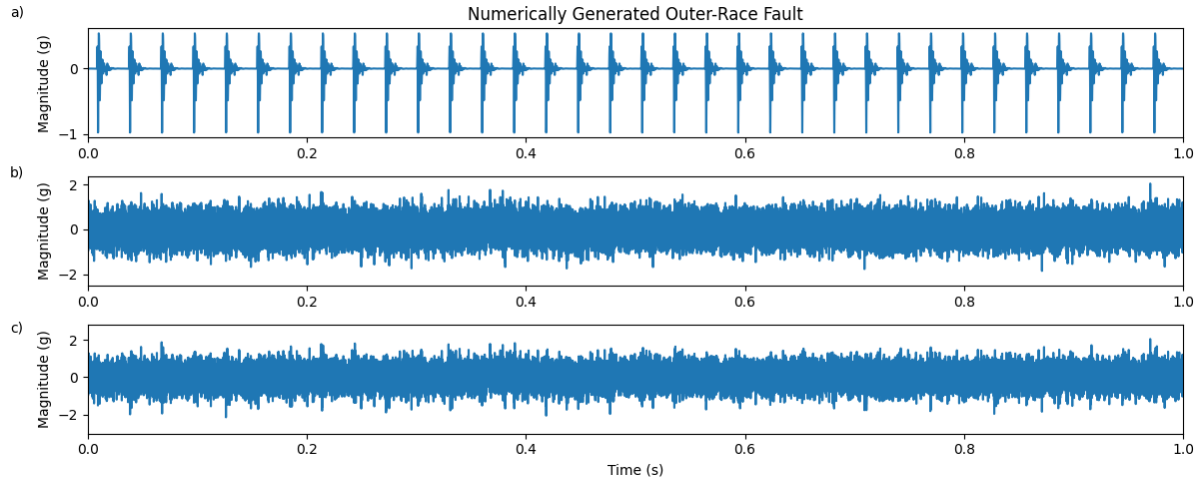


Figure 6: Example fault signal for demonstration purposes. a) Numerically generated fault signal, b) Gaussian noise (-10dB), c) Final example signal

3.2 Automatic Signal Filtering

This section defines the process that is used to automatically identify the optimal frequency band that includes the characteristic bearing fault frequencies. Once the band has been identified, the raw vibration signal is filtered around it to remove excess noise and enhance the visibility of fault-related frequencies.

3.2.1 Power Envelope Spectrum

The PES is an advanced demodulation tool [26] that combines the strengths of multiple Generalised Envelope Spectra (GES). This is done to improve the detection of cyclostationary components in noisy and impulsive vibration signals. It provides a better representation of fault-related content by combining the complementary performance of GESs computed with different transformation parameters.

The GES is defined by constructing a family of Generalised Envelopes (GEs) using a simplified Box-Cox transformation that is applied to the analytical signal. For a vibration signal ($x(t)$), the analytical signal ($a(t)$) is found using the Hilbert transform ($H\{\cdot\}$):

$$a(t) = x(t) + jH\{x(t)\} \quad (1)$$

The GE for parameter (p) is defined as:

$$GE_p(t) = \begin{cases} \ln(|a(t)|), & p = 0 \\ |a(t)|^p, & p > 0 \end{cases} \quad (2)$$

The corresponding GES is found by applying the Fourier transform ($F\{\cdot\}$) to the GE:

$$GES_p(f) = \begin{cases} F\{\ln(|a(t)|)\}, & p = 0 \\ F\{|a(t)|^p\}, & p > 0 \end{cases} \quad (3)$$

With this definition of the GES, it is possible to obtain spectra for several parameter values. Each GES will emphasise the signal differently, as the transformation parameter (p) controls how strongly the envelope spectrum responds to noise and modulation components. It is possible to combine these complementary components to form the PES, which provides a more robust representation of the characteristic fault frequencies.

The PES is defined as the element-wise product of the GES calculated for a selected set of parameters:

$$PES(f) = \prod_{m=1}^M GES_{p_m}(f) \quad (4)$$

Where p_m is the parameter that corresponds to the m th GES that is used to construct the PES.

Chen and Zhang [26] showed that the parameters $p = 0.5, 0.8, 1$ and 2 give good performance for bearing related vibration signals. This results in a PES equation of:

$$PES(f) = GES_{0.5}(f)GES_{0.8}(f)GES_1(f)GES_2(f) \quad (5)$$

Figure 7 shows the GES results for the parameters $p = 0.5, 0.8, 1$ and 2 for the demonstration vibration signal. Figure 8 shows the PES obtained by combining the GESs, illustrating how well the PES highlights fault-related features while reducing the influence of noise.

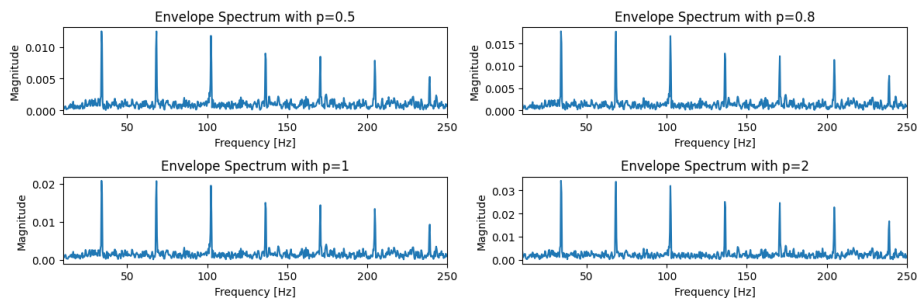


Figure 7: GESs for $p = 0.5, 0.8, 1$, and 2 (Demonstration Signal)

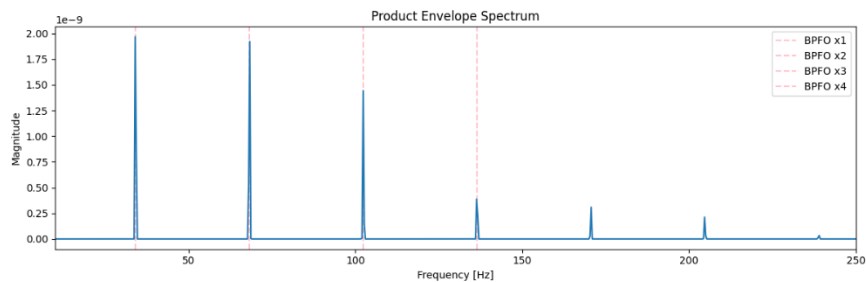


Figure 8: Resulting PES (Demonstration Signal)

3.2.2 Product Envelope Spectral Optimisation-gram

The main goal of the PESOgram developed by Chen and Zhang [1] is to automatically identify the optimal frequency band that highlights characteristic fault frequencies in a vibration signal. The PESOgram introduces an enhanced envelope analysis that evaluates the PES across a defined set of frequency bands to identify the band that maximises the fault signal-to-noise ratio.

The PESOgram assesses how strongly each frequency band highlights characteristic fault frequencies by applying the PES within a filter-band framework. The method for building the PESOgram begins by dividing the signal into a set of narrow frequency bands using the 1/3 binary tree filter bank developed by Antoni [38].

For each band in the filter bank, the signal is band-pass filtered, and the PES is calculated for the filtered signal. The diagnostic strength of each filter band is then quantified using the FDSNR, a measure of how strongly the PES emphasises the expected fault harmonics relative to the noise.

For rolling element bearings, the fault-related modulation normally appears at one of the characteristic bearing frequencies. These include the Ball Pass Frequency of the Outer race (BPFO), the Ball Pass Frequency of the Inner race (BPFI), the Ball Spin Frequency (BSF), and the Fundamental Train Frequency (FTF) [9]. These are defined as:

$$BPFO = \frac{nf_r}{2} \left[1 - \frac{d}{D} \cos(\phi) \right] \quad (6)$$

$$BPFI = \frac{nf_r}{2} \left[1 + \frac{d}{D} \cos(\phi) \right] \quad (7)$$

$$BSF = \frac{f_r D}{2d} \left[1 - \left(\frac{d}{D} \cos(\phi) \right)^2 \right] \quad (8)$$

$$FTF = \frac{f_r}{2} \left[1 - \frac{d}{D} \cos(\phi) \right] \quad (9)$$

Where n is the number of rolling elements, d is the diameter for each rolling element, and D is the pitch diameter of the bearing. f_r represents the number of rotations per second and ϕ represents the contact angle within the bearing. This project focuses on a deep groove ball bearing, so $\phi = 0$.

The FDSNR is defined as:

$$FDSNR = \frac{\frac{1}{H} \sum_{h=1}^H \max_{\alpha \in \Omega_h} \left((PES(\alpha))^2 \right)}{\frac{1}{L-H} \left[\sum_{l=1}^L (PES(f_l))^2 - \sum_{h=1}^H \max_{\alpha \in \Omega_h} \left((PES(\alpha))^2 \right) \right]} \quad (10)$$

Where H is the number of characteristic fault frequencies considered, while α represents the characteristic frequency defined as $\alpha \in \{BPFO, BPFI, BSF, FTF\}$ depending on the type of fault present. For each harmonic h , a small search region $\Omega_h = \{\alpha: |\alpha - hf_0| \leq \Delta f\}$ is defined near the expected harmonic position, where f_0 is the fundamental fault frequency and Δf is the half-width search window. The total number of frequency bins within the defined bandwidth is denoted by L , and f_l refers to the l th frequency bin within the band. Similarly to Chen and Zhang [26] H is set to a value of 3, and $f_l = 3.5\alpha$, this ensures that the characteristic frequency of interest along with its 2 harmonics are considered.

Figure 9 shows the PESOgram that has been constructed with the demonstration signal, along with the PES that corresponds to the frequency band that achieved the highest FDSNR score. This

shows how the PESOgram identifies the band in which the fault-related components are most strongly represented. Figure 10 then shows the raw demonstration signal and the signal obtained by applying a band-pass filter using the optimal band from the PESOgram. This demonstrates how effective the selected band is at highlighting the fault-related features.

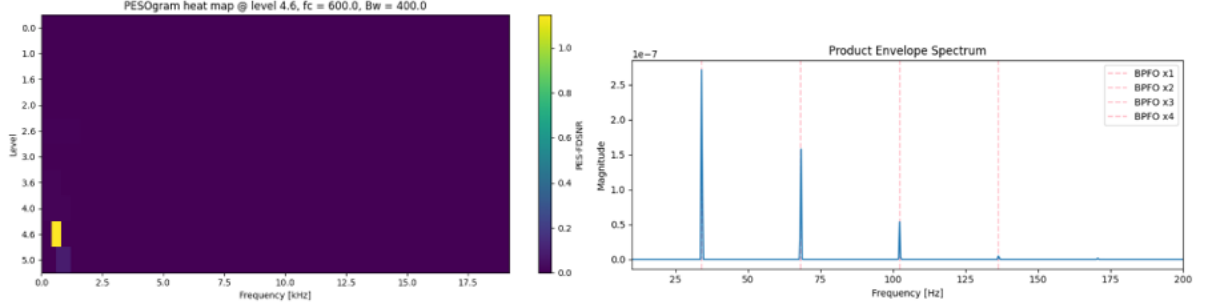


Figure 9: PESOgram (Left) and PES filtered (Right) for the demonstration signal

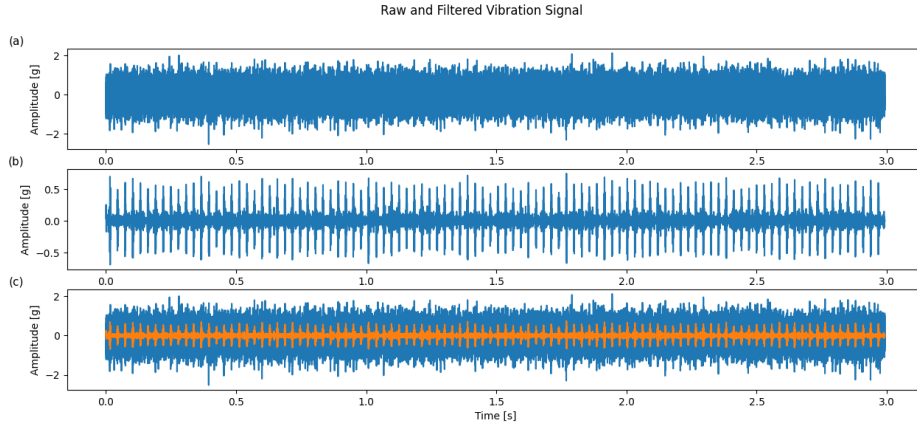


Figure 10: a) Raw demonstration signal, b) Filtered demonstration signal, c) Filtered over raw demonstration signal

3.3 Feature Extraction

This section defines the method for extracting features from the filtered vibration signals and preparing them for the next step in the detection pipeline. It begins by defining the feature set to be extracted from each accelerometer's vibration signal. Then, the construction of the feature vector that combines the relevant features from all vibration signals. Followed by the formation of a feature matrix that will contain the complete set of features from each accelerometer for every time sample.

3.3.1 Feature Set

A short list of four features has been selected for extraction from each filtered vibration signal. These features are chosen to complement the filtering process and provide reliable indicators of repeating cyclostationary impulses. The first two features are derived directly from the filtering method: the FDSNR and the magnitude of the first harmonic peak (P_1) in the PES. The FDSNR reflects the expected frequency-domain energy, while the first-harmonic magnitude is useful, as the PES emphasises fault-related components. The other two features are taken from Peeters et al. [36] and are used as sparsity measures. These features are the spectral negentropy (ΔI_E) and the $\frac{l_2}{l_1}$ - norm. These features are included because they quantify the signal's sparsity in the frequency domain. Fault-related events produce sparse spectral patterns, and these features provide a measure of how concentrated the spectral energy is.

The FDSNR is calculated using equation (10), the same equation used in the automatic signal-processing stage. The value of P_1 is found from the following equation:

$$P_1 = \max_{f \in [(1-\tau)f_m, (1+\tau)f_m]} PES(f) \quad (11)$$

Where τ is the allowable search tolerance, defined as 0.05. This will allow the first harmonic to deviate slightly from its expected harmonic position. Figure 11 shows the PES for the demonstration signal, along with the corresponding search area for P_1 included.

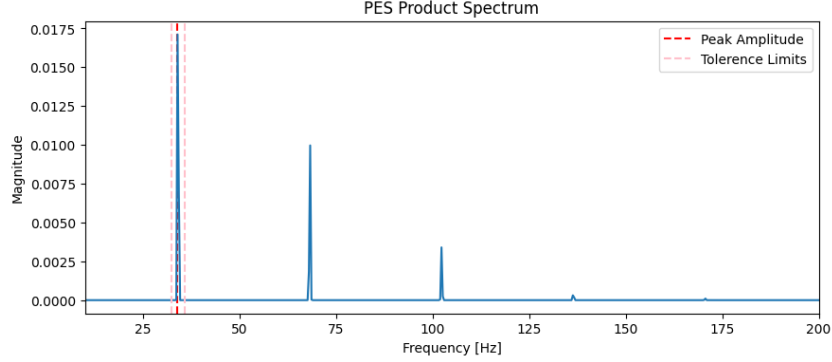


Figure 11: PES for demonstration signal with search area for P_1

Peeters et al. [36] define the spectral negentropy as:

$$\Delta I_E = \left\langle \frac{|E(f)|^2}{\langle |E(f)|^2 \rangle} \ln \left(\frac{|E(f)|^2}{\langle |E(f)|^2 \rangle} \right) \right\rangle \quad (12)$$

Where $E(f)$ represents the envelope spectrum of the analytical signal ($E(f) = F\{|a(t)|\}$). They define the $\frac{l_2}{l_1}$ - norm as:

$$\frac{l_2}{l_1} = \frac{\sqrt{\sum_{n=1}^N |E(f)|^2}}{\sum_{n=1}^N |E(f)|} \quad (13)$$

Where n is the point number and N is the total number of points in each signal.

3.3.2 Feature Vector and Feature Matrix

For each time interval, the four vibration features discussed above are extracted from each accelerometer. These features are then grouped together to form a feature vector for the accelerometer (A), denoted as $\mathbf{h}_A$. An important note: all vectors and matrices are represented in bold to distinguish them from scalars. For a given accelerometer, the feature vector for an interval is expressed as:

$$\mathbf{h}_A = \left[FDSNR_A, (P_1)_A, (\Delta I_E)_A, \left(\frac{l_2}{l_1} \right)_A \right] \quad (14)$$

The complete feature vector ($\mathbf{H}$) is constructed by concatenating the feature vectors from each relevant accelerometer. Each accelerometer signal is separated into multiple time samples (s), allowing for feature vectors to be extracted from different points of a given signal. For example,

when three accelerometers are being analysed, the complete feature vector for each time sample will include $4a$ features, where a is the total number of accelerometers used. This method allows the use of as few or as many accelerometer signals as needed, as long as there is sufficient data to train the model. For 3 accelerometers, the complete feature vector will look like:

$$\mathbf{H} = [\mathbf{h}_{A_1}, \mathbf{h}_{A_2}, \mathbf{h}_{A_3}] \quad (15)$$

The feature vectors from all time samples are assembled into a feature matrix ($\mathbf{X}$) that represents the full set of extracted features. The feature matrix for a signal will look like:

$$\mathbf{X} = \begin{bmatrix} \mathbf{H}_1 \\ \mathbf{H}_2 \\ \vdots \\ \mathbf{H}_s \end{bmatrix} \quad (16)$$

3.4 Feature Weighting

This is the point in the pipeline where it becomes important to look at the purpose of the data. If the input vibration signals are healthy and intended for training the fault detection model, the feature matrix is denoted as $\mathbf{X}_{train}$. If the input signals are not intended to train the model, the feature matrix is denoted as $\mathbf{X}_{test}$. To ensure that all the features from a feature vector contribute equally, and to prevent any single feature that has a larger magnitude from dominating the others, the feature vectors must be normalised, so they all operate on the same bounds. The feature weighting ($\mathbf{W}$) matrix used for this normalisation is constructed from only the training feature matrix. The method for calculating the weighing matrix and the weighted matrices follows the approach described by Chaleshtori and Aghaie [37]. The weighted matrix is calculated as follows:

$$\mathbf{W} = \begin{bmatrix} \frac{1}{\sigma_1} & 0 & 0 & 0 \\ 0 & \frac{1}{\sigma_2} & 0 & 0 \\ 0 & 0 & \ddots & \vdots \\ 0 & 0 & \dots & \frac{1}{\sigma_j} \end{bmatrix} \quad (17)$$

Where σ_j is the standard deviation for the j th training feature. With this, the weighted training matrix ($\mathbf{S}_{train}$) and the weighted test matrix ($\mathbf{S}_{test}$) are calculated from the following equations:

$$\mathbf{S}_{train} = \mathbf{W}^T \mathbf{X}_{train}^T \mathbf{X}_{train} \mathbf{W} \quad (18)$$

$$\mathbf{S}_{test} = \mathbf{W}^T \mathbf{X}_{test}^T \mathbf{X}_{test} \mathbf{W} \quad (19)$$

3.5 Dimensional Reduction Using PCA

PCA is useful for this application because it reduces the dimensionality of the feature space while preserving the most important variation in the data. This helps remove redundancy among correlated features, improving computational efficiency in the subsequent detection stages. By

projecting the data onto a smaller set of principal components, PCA preserves most of the important structure of the vibration features while simplifying the data representation [37].

The weighted feature matrix is first used to find the covariance structure. PCA then finds the directions in a new coordinate space where the data varies the most. These directions form an orthogonal set of vectors that is known as the loading matrix ($\mathbf{Z}$). The original weighted features are then projected onto this new coordinate system to produce a list of principal component scores. The transformed weighted feature matrix is denoted as $\mathbf{T}$ and can be described as:

$$\mathbf{T} = \mathbf{Z}\mathbf{S} \quad (20)$$

To find $\mathbf{Z}$, the covariance matrix of $\mathbf{S}_{train}$ is decomposed into its eigenvalues and eigenvectors. The eigenvectors define the principal components, and the associated eigenvalues indicate how much variance each component explains. The principal components are then ranked in descending order of variance, ensuring that the first principal component contains the most variance, with variance decreasing for each subsequent component.

As with the feature weighting process, the PCA is fitted using only the weighted training matrix. The training transformed weighted matrix is denoted as $\mathbf{T}_{train}$ and the test transformed weighted matrix is denoted as $\mathbf{T}_{test}$. When the transformation is applied to weighted data, the method is referred to as Weighted PCA (WPCA).

3.6 The Gaussian Mixture Model

After the weighted feature matrix has been transformed into the principal component space $\mathbf{T}_{train}$ is used to train the GMM. The GMM models the distribution of healthy vibration data by representing it as a mixture of multiple Gaussian Components. Each component captures a different region of the transformed data.

3.6.1 Methods for Model Stability

To prevent the model from overfitting to the training data, several measures have been taken. The first is to use a diagonal covariance structure. As described by Murphy [39], restricting the covariance matrix in this way will limit the number of parameters that will need to be estimated. In vibration analysis, healthy signals often exhibit some variation, especially when random impulses are present. These impulses may deviate the signal from the learned healthy behaviour, but if healthy, it will still need to be classified correctly. By using a diagonal covariance matrix, the model is prevented from becoming overly sensitive to small deviations in the feature space. This helps limit natural vibration fluctuations and random impulses from being misclassified as faults.

Another method to help against overfitting is to limit the number of Gaussian Mixture components (K) when there is insufficient data. To ensure that there is enough data to properly estimate each component, an upper bound is applied to K . The maximum allowable value of K is approximated as:

$$K_{max} \approx \frac{N_H}{10(2d_H + 1)} \quad (21)$$

Where N_H is the number of feature vectors used to train the model, and d_H is the dimensionality of the PCA-transformed feature vector. This expression is not an exact limit, but rather a guideline inspired by Murphy's [6] observations that stable covariance estimations require the number of

samples to be significantly more than the number of free parameters. The factor of 10 is intended to provide a conservative margin to maintain the model's stability.

3.6.2 Selection of the Number of Mixture Components

The value of K for the GMM is not chosen manually. Rather, it will be determined automatically using the Bayesian Information Criterion (BIC). BIC provides a balance between model fit and the model's complexity by penalising models with a higher number of parameters [40]. This makes it suitable for selecting K , as it helps to prevent overfitting and favours models that explain the data well.

For each candidate value of K , a GMM is fitted to the PCA-transformed matrix $\mathbf{T}_{train}$ and the corresponding BIC value is calculated. The BIC curve often decreases rapidly in the beginning and then starts to stabilise, with small improvements as K increases. The goal is to use the minimum number of mixture components that explain the data well. The selection process will check whether the next BIC value yields a significant decrease. If the improvement in BIC values is minimal and the current BIC value is close to the minimum, then the current K value is selected to train the model.

This approach will ensure that the final model will be both compact and well-fitted to the training data. If it is found that the model does not stabilise at any point, the model with the lowest overall BIC is selected.

3.6.3 GMM Use for Fault Detection

After the optimal number of mixture components has been selected and the GMM has been trained on $\mathbf{T}_{train}$, it will then be used to evaluate unseen vibration data. Each incoming sample is one that has followed the same filtering, feature extraction, and WPCA transformation ($\mathbf{T}_{test}$) is then passed to the GMM to calculate its Negative Log-Likelihood (NLL) score. Samples that are close to the learned healthy distribution will receive a low NLL score, while samples that deviate from the learned healthy distribution will receive a high NLL score. A detection threshold is set based on the distribution of training NLL scores, and any sample that exceeds this threshold will be flagged as an anomaly and a potential fault.

3.7 Model Performance

The overall detection model will be evaluated based on its ability to correctly classify whether a sample includes a fault. To do this, the accuracy metric from Chaleshtori and Aghaie [37] will be used. This metric is defined as:

$$accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (22)$$

Where TP is the number of true positive results, TN is the number of true negative results, FP is the number of false positive results, and FN is the number of false negative results.

4 Numerical Bearing Model

A two-dimensional lumped-parameter model based on a 5-degree-of-freedom (DOF) model developed by Randall and Sawalhi [12] is used to simulate the vibration response of a rolling-element bearing with an outer-race defect. The model combines nonlinear Hertzian contact forces with a spring-mass-damper representation of the bearing housing and test bench, enabling a realistic simulation of both bearing dynamics and the transmission paths to multiple accelerometers. The model used in this research is an adapted 8-DOF model. The additional DOFs include extra resonators and transmission paths compared to the 5-DOF model. Figure 12 shows the base 5-DOF model from Randal and Sawalhi on the left and the updated 8-DOF model on the right.

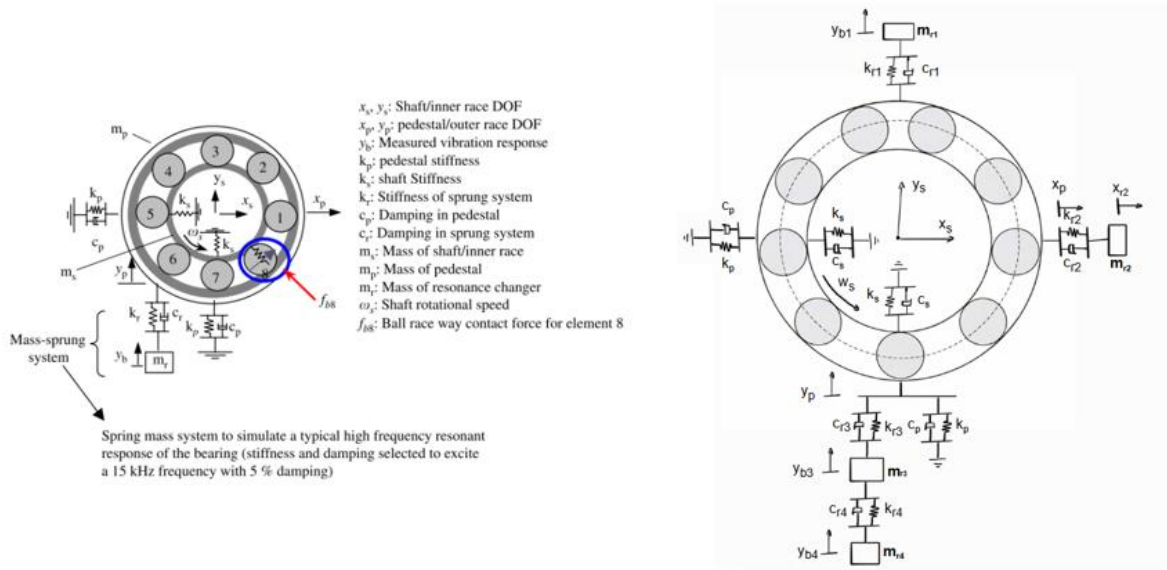


Figure 12: 5-DOF bearing model [12] (left), 8-DOF bearing model (right)

m_{r1}, m_{r2} , and m_{r4} represent the masses of the virtual accelerometers, they are accompanied by their respective spring constants (k_{r_i}) and damping ratios (c_{r_i}). This model represents a setup with one accelerometer at the top of the bearing, aligned with the y-axis, one on the side, aligned with the x-axis, and one on the base of the test bench. With this setup, it is possible to measure horizontal and vertical vibrations from the fault with ideal transmission paths, while the accelerometer on the base will help visualise how the vibration signal propagates to a less-ideal mounting location. The positions of the virtual accelerometers will show the influence of accelerometer orientation as well as of transmission paths.

4.1 Structural Model and Equations of Motion

The dynamics of the 8-DOF LPM are shown in Figure 12 (right) are expressed as a system of coupled second-order differential equations. These equations control the motion of the bearing components and the structural transmission paths at each time step of the simulated vibration signal. The inner race and shaft assembly is modelled with two translational degrees of freedom (x_s, y_s) and the outer race and pedestal are also modelled with two translational degrees of freedom (x_p, y_p). The other four degrees of freedom represent the structural transmission path and the virtual accelerometers:

- y_{b1} : vertical DOF of virtual accelerometer mass (Top of bearing housing)

- x_{b_2} : Horizontal DOF of virtual accelerometer mass (Side of bearing housing)
- y_{b_3} : vertical DOF of the intermediate transmission path mass
- y_{b_4} : Vertical DOF of virtual accelerometer mass (Base mounted)

These components form an 8-DOF system that captures the dynamics and relative vibration signals for each virtual accelerometer. The second-order differential equations that describe the inner race/shaft subsystem are:

$$m_s \ddot{x}_s + f_x + c_s \dot{x}_s + k_s x_s = 0 \quad (22)$$

$$m_s \ddot{y}_s + f_y - 9.81m_s + c_s \dot{y}_s + k_s y_s = 0 \quad (23)$$

Where f_x and f_y represent the nonlinear Hertzian contact forces from the rolling elements. The $9.81m_s$ term represents the gravitational force acting on the system; it is the only external force included in the model. The dot notation is used to denote time derivatives for position-based variables. A single dot above a variable denotes the first derivative in time (velocity), and a double dot denotes its second derivative (acceleration).

The second-order differential equations that will describe the outer race/pedestal are:

$$m_p \ddot{x}_p - f_x + \dot{x}_p(c_p + c_{r_2}) + x_p(k_p + k_{r_2}) - c_{r_2} \dot{x}_{b_2} - k_{r_2} x_{b_2} = 0 \quad (24)$$

$$m_p \ddot{y}_p - f_y + \dot{y}_p(c_s + c_{r_1} + c_{r_3}) + y_p(k_s + k_{r_1} + k_{r_3}) - c_{r_1} \dot{y}_{b_1} - c_{r_3} \dot{y}_{b_3} - k_{r_1} y_{b_1} - k_{r_3} y_{b_3} = 0 \quad (25)$$

The equation that will describe the excited mass representing the transmission is:

$$m_{r_3} \ddot{y}_{b_3} + c_{r_3}(\dot{y}_{b_3} - \dot{y}_p) + k_{r_3}(y_{b_3} - y_p) + c_{r_4}(\dot{y}_{b_3} - \dot{y}_{b_4}) + k_{r_4}(y_{b_3} - y_{b_4}) = 0 \quad (26)$$

Finally, the second-order differential equations that will describe the motion of the virtual accelerometer masses are:

$$m_{r_1} \ddot{y}_{b_1} + c_{r_1}(\dot{y}_{b_1} - \dot{y}_p) + k_{r_1}(y_{b_1} - y_p) = 0 \quad (27)$$

$$\ddot{x}_{b_2} + c_{r_2}(\dot{x}_{b_2} - \dot{x}_p) + k_{r_2}(x_{b_2} - x_p) = 0 \quad (28)$$

$$m_{r_4} \ddot{y}_{b_4} + c_{r_4}(\dot{y}_{b_4} - \dot{y}_{b_3}) + k_{r_4}(y_{b_4} - y_{b_3}) = 0 \quad (29)$$

4.2 Bearing Kinematics and Contact Mechanics

In order to calculate the nonlinear bearing forces (f_x and f_y) that appear in the equations of motion, the following will need to be clearly defined:

- Motion of the rolling elements
- Contact deformation between races

- The Hertzian contact forces

This section will introduce these components and clearly define all of the variables.

4.2.1 Cage speed

The rotational speed of the bearing cage (ω_c) determines how fast the rolling elements circulate around the bearing. This is defined as:

$$\omega_c = \left(1 - \frac{d}{D}\right) \left(\frac{\omega_s}{2}\right) \quad (30)$$

Where d is the diameter of the rolling elements, D is the pitch diameter, and ω_s is the rotational speed of the shaft/inner race.

4.2.2 Angular Position of Each Rolling Element

The angular position of the j -th rolling element ($j = 1, 2, \dots, n_b$), determines where each ball will be located around the bearing at all time steps. The angular position [41] for the j -th ball (ϕ_j) is defined as:

$$\phi_j = \frac{2\pi(j-1)}{n_b} + \omega_c dt + \phi_0 + \phi_{slip} \quad (31)$$

Where n_b is the number of rolling elements, dt is the time, ϕ_0 is the previous cage position, and ϕ_{slip} is the phase deviation of the angular position due to slip between components.

This ensures that each rolling element maintains its correct spacing and relative position as it rolls within the bearing.

4.2.3 Outer Race Fault Switch

The position of an outer race defect is fixed in space. A smooth switching function (β_j) is used in the model to add clearance when the rolling element passes over the defect. The smooth function will allow us to create a more realistic fault signature. The switching function used:

$$\phi_{rel,j} = (\phi_j - \phi_d)2\pi \quad (32)$$

$$\beta_j = \begin{cases} 0.5 \left[1 - \cos\left(2\pi \frac{\phi_{rel,j}}{\Delta\phi_d}\right) \right], & \delta_j > 0 \\ 0, & otherwise \end{cases} \quad (33)$$

4.2.4 Contact deformation

The relative deformation between the inner and outer race in the direction of the j -th rolling element is [41]:

$$\delta_j = (x_s - x_p)\cos\phi_j + (y_s - y_p)\sin\phi_j - c - \beta_j C_d \quad (34)$$

Where c is the element clearance and C_d is the depth of the fault.

As the elements can only provide a force when they are in contact with both the inner and outer race, a contact switch [41] (γ_j) is defined as:

$$\gamma_j = \begin{cases} 1, & \delta_j > 0 \\ 0, & \text{otherwise} \end{cases} \quad (35)$$

4.2.5 Hertzian Contact Forces

It is now possible to use the Hertzian contact theory to generate the resulting force. The resulting force is found from summing the individual x and y force components from each rolling element to obtain a total non-linear, time-varying bearing force. The bearing forces [41] are defined as:

$$f_x = k_b \sum_{j=1}^{n_b} \gamma_j \delta_j^{1.5} \cos \phi_j \quad (36)$$

$$f_y = k_b \sum_{j=1}^{n_b} \gamma_j \delta_j^{1.5} \sin \phi_j \quad (37)$$

Where k_b is the Hertzian contact stiffness.

4.3 Numerical Integration and Data Generation

The 8-DOF system is represented in first-order state space form. The state vector ($z(t)$) contains the displacements and velocities:

$$z(t) = [x_s, \dot{x}_s, y_s, \dot{y}_s, x_p, \dot{x}_p, y_p, \dot{y}_p, y_{b_1}, \dot{y}_{b_1}, x_{b_2}, \dot{x}_{b_2}, y_{b_3}, \dot{y}_{b_3}, y_{b_4}, \dot{y}_{b_4}] \quad (38)$$

This system is then integrated numerically according to:

$$\dot{z}_t = f(z(t), t) \quad (39)$$

Where f is obtained from the equations of motion.

The extracted accelerometer signals for each accelerometer are as follows:

$$A_1(t_k) = \ddot{y}_{b_1}(t_k), A_2(t_k) = \ddot{x}_{b_2}(t_k), A_3(t_k) = \ddot{y}_{b_4}(t_k) \quad (40)$$

Each time step from t_k to t_{k+1} follow this path:

1. Update rolling element positions: Equation (31)
2. Compute contact deformation: Equation (34)
3. Apply the defect switch: Equation (35)
4. Compute Hertzian forces: Equation (36-37)
5. Evaluate $z(t)$: Equation (38)
6. Integrate to the next time step for each differential with Runge-Kutta ODE solver: Equation (22-29)

7. Extract accelerometer signals: Equation (40)

4.4 Generation of Vibration Signals

For the numerical vibration signal, the main goal is to assess how well the detection pipeline handles non-impulsive operating conditions. This is important because the detection model must reliably function under both impulsive and standard operating conditions. The aim is to determine the point at which a developing fault becomes detectable and to find the model's sensitivity limits under non-impulsive conditions.

The bearing vibration signal represents a bearing rotating at 500 RPM, with an outer-race fault located at the bottom of the bearing in the negative y-direction. All variables describing the model follow those used by Randal and Sawalhi [1]. The mass m_{r_3} is selected to be larger than the masses that represent the rotating shaft and pedestal. This is an intentional decision, as the transmission path is intended to reflect a less-than-ideal case in which extra energy is required for the vibration to propagate.

The bearing dimensions are chosen to match those of the 6210 deep-groove ball bearing used during the experimental phase. All of the values that differ from the values defined by Randal and Sawalhi [1] are shown in Table 3. The purpose of the vibration model is not to replicate the exact dynamic behaviour expected during testing, but rather to provide a representative signal shape for evaluating the detection pipeline. This means the simulated and experimental results are not directly comparable. However, both datasets will provide meaningful insights into the detection model's limits and capabilities.

Table 3: Bearing Dimensions

Parameter	Value	Unit
n	10	-
d	12.7	mm
D	70	mm
m_{r_3}	20	kg

4.4.1 Description of Generated Vibration Signals

Figure 13 shows a half-second segment of the simulated healthy vibration signal. Plots a, c, and e, present the time-domain responses for the accelerometers A1, A2, and A3, while plots b, d, and f show the corresponding envelope spectra. The time-domain responses are consistent among all three sensors, indicating that the model captures the system's dynamics well. The magnitudes of the responses from accelerometers A1 and A2 are similar, as expected: there is no fault present, so they should both measure the same amount of vibration energy. The response from A3 is much lower, which aligns with expectations given the transmission path that requires more energy.

In the envelope spectra, the main peaks align closely with the characteristic frequency of an outer race. Even though the simulated vibration signal represents a healthy case, small variations in loading and geometry are likely producing components near the expected frequency. This, along with the fact that the simulated signal contains no noise, means that the low-level healthy response is not masked by noise and can resemble an outer-race fault. When random noise is added, these small components would typically be overwhelmed.

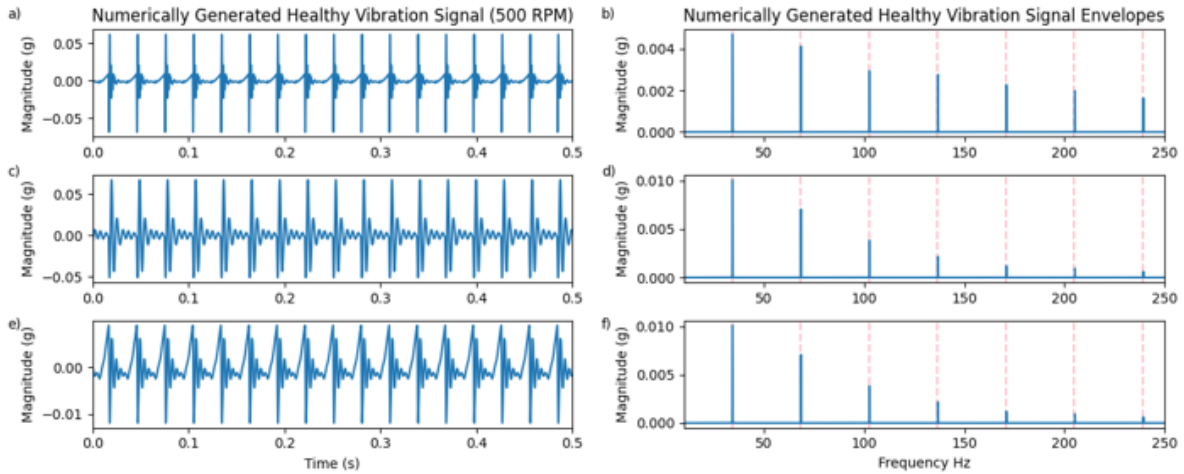


Figure 13: Simulated healthy vibration signals for 500RPM: a) A1 time response, b) A1 envelope spectrum, c) A2 time response, d) A2 envelope spectrum, e) A3 time response, f) A3 envelope spectrum

Figure 14 shows the simulated vibration signal for the case with an outer-race fault. The layout follows that of Figure 13. The time-domain signal clearly exhibits periodic events followed by high-frequency resonance, which is characteristic of an outer-race fault.

When comparing the magnitudes of the responses from each accelerometer with those of its healthy counterpart, all show a noticeable increase, as expected. Accelerometer A1 is aligned with the fault, mounted close to it, and in the same line of action as the gravitational load, so it experiences the largest increase, as expected. Accelerometer A2 is not aligned with the fault, so even though the magnitude increases, the overall change is less. Accelerometer A3 shows a larger increase than A2 when compared to their healthy cases, indicating that its orientation and transmission path all contribute more of the fault energy to reach it.

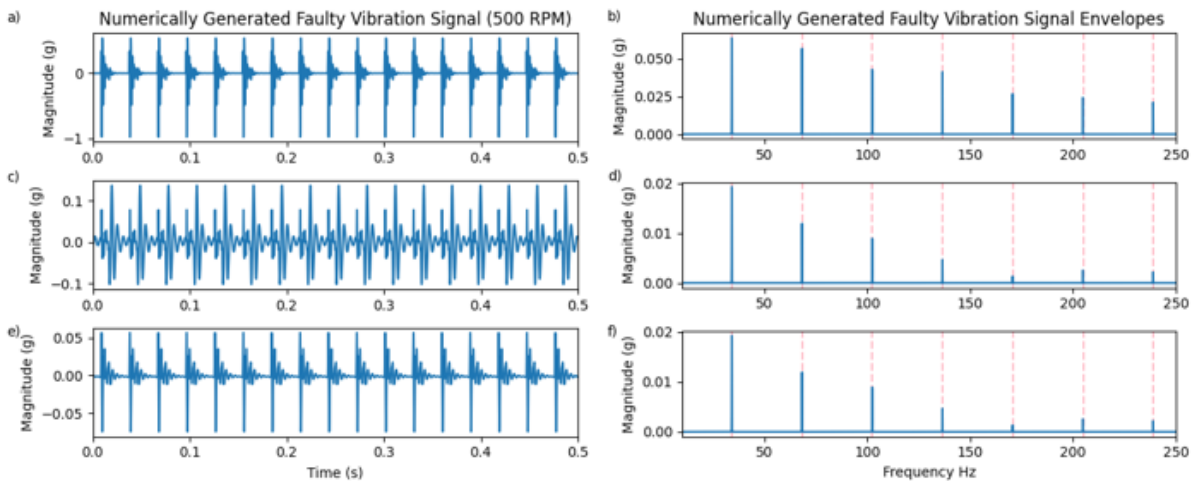


Figure 14: Simulated faulty vibration signals for 500RPM: a) A1 time response, b) A1 envelope spectrum, c) A2 time response, d) A2 envelope spectrum, e) A3 time response, f) A3 envelope spectrum

4.4.2 Adding Noise to the Signals

Gaussian noise is added to the vibration signal to assess the extent to which it affects detection performance. The noise level is selected so that the simulated healthy signal will have a signal-

to-noise ratio of -35 dB. This value is based on findings from Peeters et al. [2], who showed that many indicators exhibit bearing-related behaviour in the range of -30 dB to -40 dB.

Since the goal is not to test the model with purely Gaussian noise, the noise level is chosen to ensure that some bearing dynamics remain in the signal. This ensures that the signal will still include some meaningful vibration content.

The equation used to determine the signal-to-noise (SNR) ratio for the signal is:

$$SNR = 10 \log_{10} \left(\frac{\sigma_f^2}{\sigma_N^2} \right) \quad (41)$$

Where σ_f is the standard deviation of the signal and σ_N is the standard deviation of the noise. The equation will be rearranged to find σ_N for each accelerometer so that the fault signal can have the same noise applied. The noisy healthy vibration signal ($x_{h_n}(t)$) is defined as:

$$x_{h_n}(t) = x_h(t) + n(t) \quad (42)$$

Where $x_h(t)$ is the simulated healthy signal and $n(t)$ is the zero-mean Gaussian noise with a standard deviation of σ_N where the noisy healthy vibration signal has an SNR of -35 dB.

Figure 15 shows the Gaussian noise that will be applied to each of the virtual accelerometer signals to ensure that they each have an SNR of -35 dB.

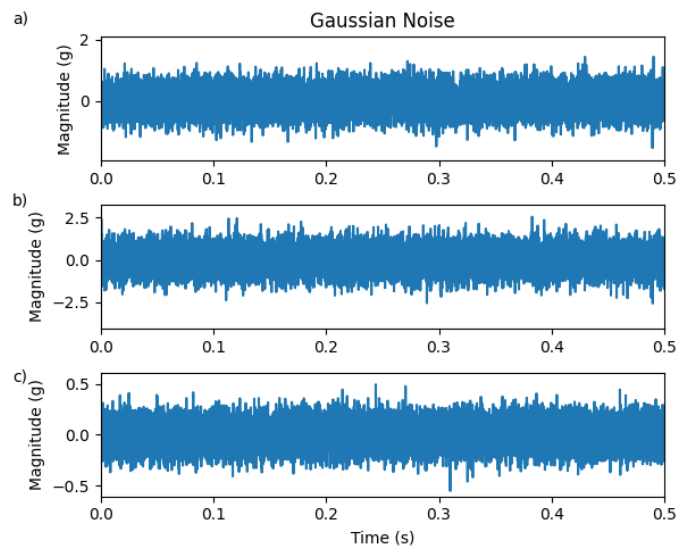


Figure 15: Gaussian noise ($n(t)$) for each accelerometer

To vary the SNR of the vibration signal that includes the fault, the noise level cannot be changed from that used for the healthy case. The noise term ($n(t)$) for each term is therefore kept constant, this means that the noise added to healthy A1 will be the same as the noise added to faulty A1 and so on. To achieve different SRN values, the magnitude of the faulty vibration signal must be adjusted instead. The noisy and scaled fault signal ($x_{f_n}(t)$) is defined as:

$$x_{f_n}(t) = kx_f(t) + n(t) \quad (43)$$

Where k is the scaling factor and $x_f(t)$ is the simulated fault signal.

Figure 16 shows the noisy healthy accelerometer signals on the left, where plots a, b and c represent accelerometers A1, A2 and A3, respectively. The right-hand side shows the noisy faulty vibration signals, where plots d, e and f correspond to A1, A2 and A3.

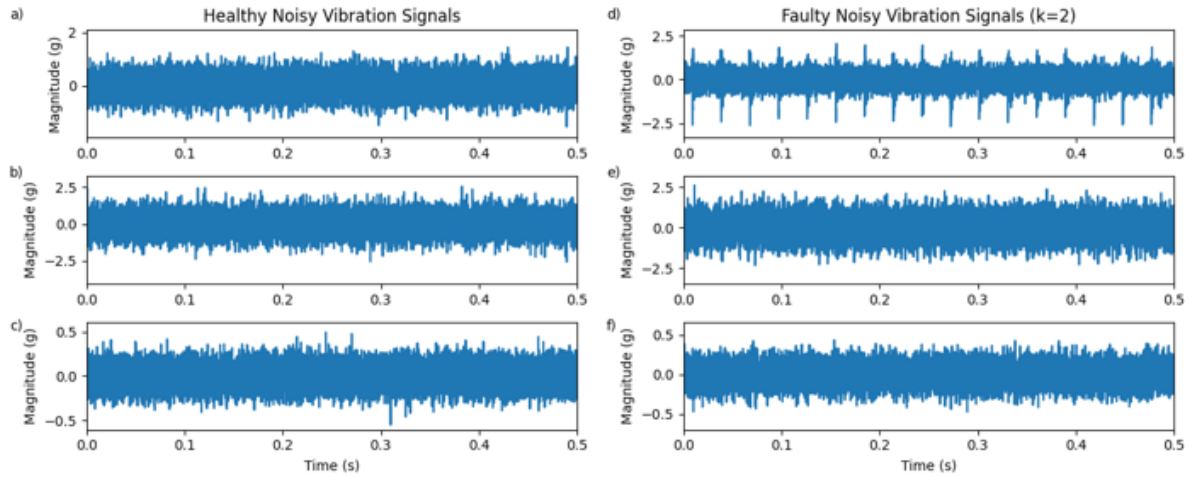


Figure 16: Noisy Vibration Signals: a) A1 Healthy with noise, b) A2 Healthy with noise, c) A3 Healthy with noise, d) A1 Faulty with noise, e) A2 Faulty with noise, f) A3 Faulty with noise

4.5 Applying Fault Detection Pipeline to Simulated Signals

The previously defined detection pipeline is applied here to the data generated by the numerical bearing model. The detection model that is described in this section is trained only on 500 RPM features. This section outlines how the PES-based filtering works and how the GMM stages behave when applied to simulated signals. The filtering results show how the PES method responds to signals with different signal-to-noise ratios. The GMM results show how the model selects the number of mixture components for the simulated feature set.

4.5.1 Filtering

As the filtering method relies on maximising the FDSNR, it is important to select a sample time that will produce a strong separation between healthy and faulty signals. Figure 17 shows the FDSNR values for several tests from the same 500 RPM case, which includes both healthy and faulty signals. The faulty signals reach a maximum FDSNR at roughly 2 seconds, where the healthy signal remains essentially at zero across all sample times. This indicates that a 2-second sample time is ideal when analysing data that is known to be at 500 RPM.

Due to the high computational costs, only the first 2 seconds of each accelerometer signal are used to determine the optimal filter band. The selected band will form the initial segment, which will then be applied to the full vibration signal. Ideally, each 2-second segment of the entire signal will be filtered, and the band that yields the best performance will be selected for the signal. This, however, was not feasible for this study due to the large number of tests, each with multiple accelerometers. It would be practical in a real-world application where the detection model is already trained, and the necessary accelerometer combinations are already defined.

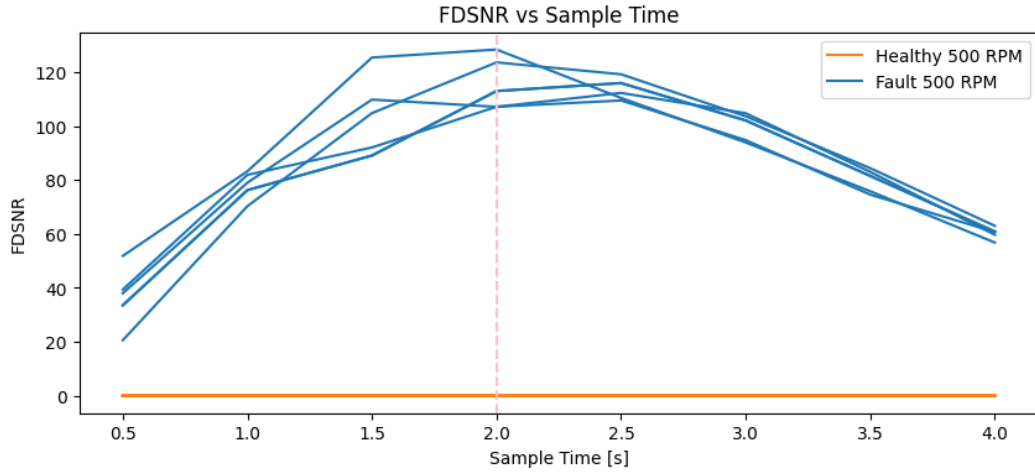


Figure 17: FDSNR vs Sample Time for 500 RPM vibration signals

Figure 18 and Figure 19 both show filtered vibration signals and use the same layout. Panels a, c, and e show the first 1 second of the raw (blue) and filtered (orange) signals from accelerometers A1, A2, and A3. Panels b, d, and f show the corresponding PES results from the filtered signal.

Figure 18 shows filtering results for a healthy bearing. The filtering process does not artificially generate frequency content near the expected fault frequency. This is also reflected in the PES plots, which do not align with the expected harmonic structure of a fault.

Figure 19 shows a signal for a case with a large magnitude fault. For these vibration signals, the filtering method successfully isolated the fault-related components from the surrounding noise. For A1, the fault signature is visible in the raw time domain signal, so the filtered result is expected. For A2 and A3, the fault signatures are heavily obscured by noise, yet the filtering process still isolates the correct frequency bands that highlight them. The corresponding PES plots all closely match the expected harmonic pattern and show significantly higher magnitudes than the healthy case.

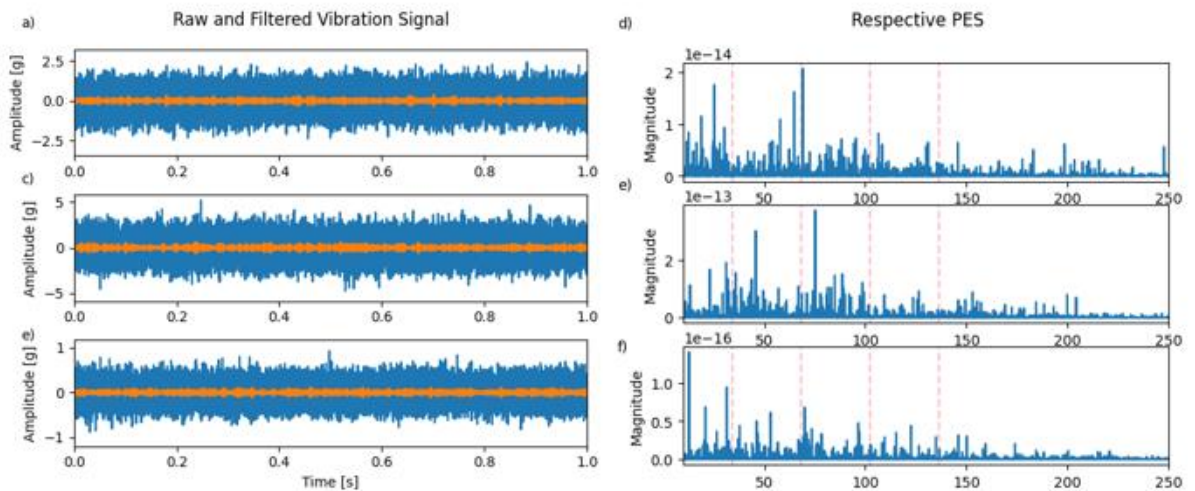


Figure 18: Segment of Healthy vibration signal for each accelerometer, a) A1 raw and filtered signal, b) PES for A1 filtered signal, c) A2 raw and filtered signal, d) PES for A2 filtered signal, e) A3 raw and filtered signal, f) PES for A3 filtered signal

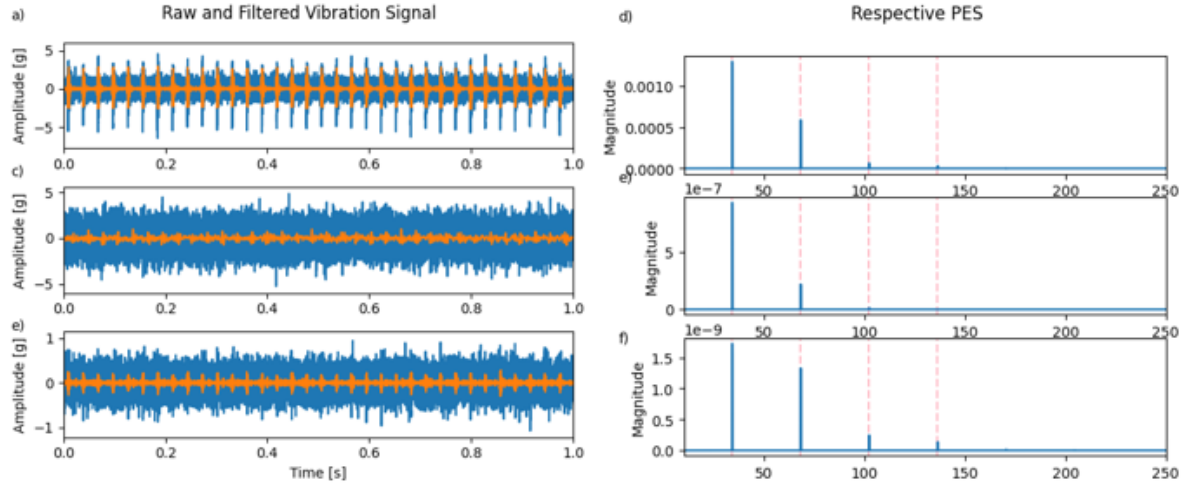


Figure 19: Segment of Faulty vibration signal for each accelerometer, a) A1 raw and filtered signal, b) PES for A1 filtered signal, c) A2 raw and filtered signal, d) PES for A2 filtered signal, e) A3 raw and filtered signal, f) PES for A3 filtered signal

4.6 GMM training

When training the GMM, the sample time is less critical than in the filtering stage. After feature weighting and PCA, only the relative differences between the healthy and faulty faults will influence the model. The GMM is also trained on multiple features rather than just a single FDSNR value. Maximising sample time for a single feature does not guarantee improved performance, as it may degrade other features. As the GMM benefits from having more data points, the sample time is reduced to 1-second segments. This doubles the number of available feature vectors for the GMM. This is particularly important when using three accelerometers, as the resulting feature space will be inherently high-dimensional.

For the numerical GMM model, 38 healthy 20-second signals were generated, giving a total of 760 one-second segments for training. Figure 20 shows the GMM training results for an accelerometer combination, A2 and A3. The PC plots and 2D GMM plots for all relevant accelerometer combinations are included in Appendix D. The left of Figure 20 shows the selection of the number of mixture components using BIC. Although the number of mixture components is often limited to 10 to avoid overfitting, the limit for this case is reduced by equation (21) when the dimensionality is too high to safely train a model with too many mixture components. For this accelerometer pair, the limit is 7. The 2D GMM plot on the right of Figure 20 shows how the GMM clusters the PCA-reduced features. The multiple groups are likely due to small variations in the bearing model initialisation, which lead to small differences in the simulated bearing dynamics.

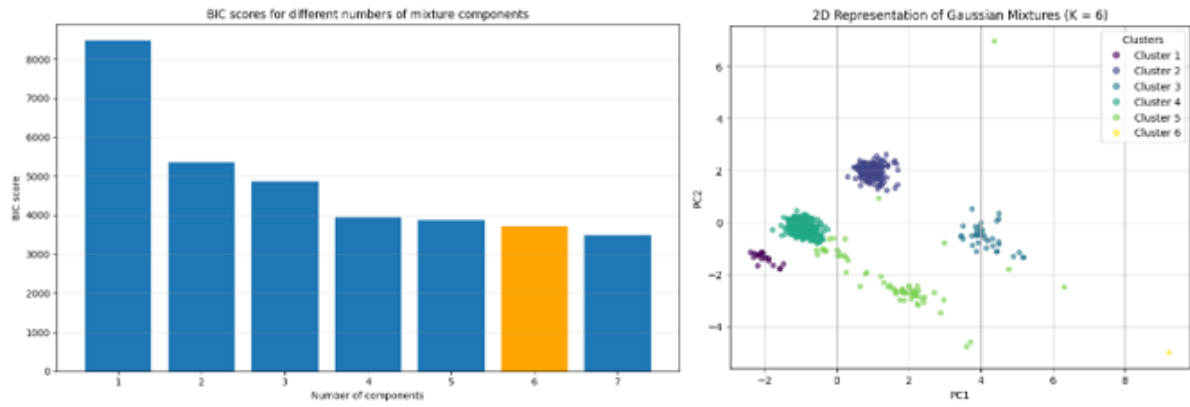


Figure 20: GMM training for accelerometer combination A2 and A3, BIC mixture selection (left), 2D representation of trained GMM (right)

4.7 Detection Results

Figure 21 provides a clear visual example of how the GMM identifies a fault. The PCA plot on the left shows a healthy test sample that clusters closely with the learned healthy distribution. These points would receive low negative log-likelihood values and therefore would not trigger an anomaly flag. In contrast, the PCA plot on the right shows the plots for data that contains a fault. None of these points lies near the healthy distribution, indicating they would likely receive a high negative log-likelihood score, and the model will likely flag this test signal as indicating a fault.

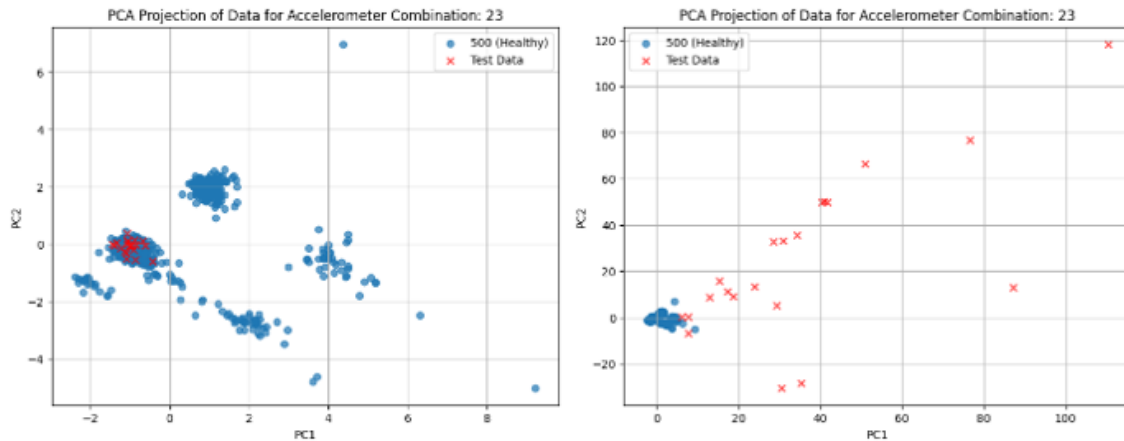


Figure 21: PC plot with healthy test data (left), PC plot with faulty test data (right)

In Figure 22, the detection accuracy of each accelerometer is shown at different noise levels. The plot indicates that Accelerometer A3 performs best at the highest noise levels, likely because the noise surrounding A3 is lower, making it easier to detect fault signals. Accelerometer A1 comes next, performing as expected in a relatively ideal scenario. Meanwhile, Accelerometer A2 is more sensitive to noise and only detects faults effectively at lower noise levels (around -20 dB), whereas A1 maintains accuracy down to about -25 dB and A3 down to about -30 dB.

In Figure 23 the fault detection accuracy versus the scaling factor for different accelerometer combinations is examined, both individually and in combination. The results from this plot show that A1 is the most responsive and can detect even the smallest faults. A3 becomes more effective as the fault grows stronger, while A2 is the slowest to respond to the fault. This shows that alignment with the fault and the direction of the force is more important than mere proximity. This plot also shows that, while operating in normal Gaussian noise, using multiple

accelerometers performs slightly worse than the best sensor in the combination; however, it still provides a balanced and reliable approach.

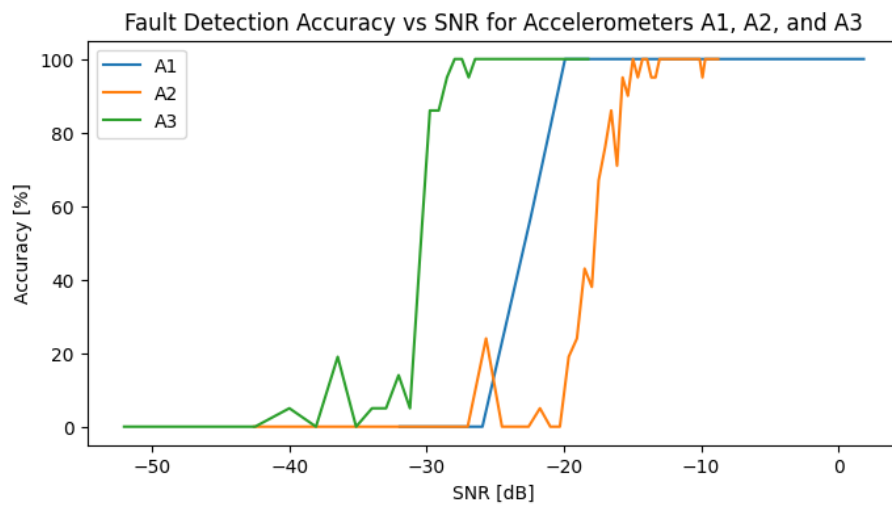


Figure 22: Plot of fault detection accuracy vs SNR for individual accelerometers A1, A2, and A3

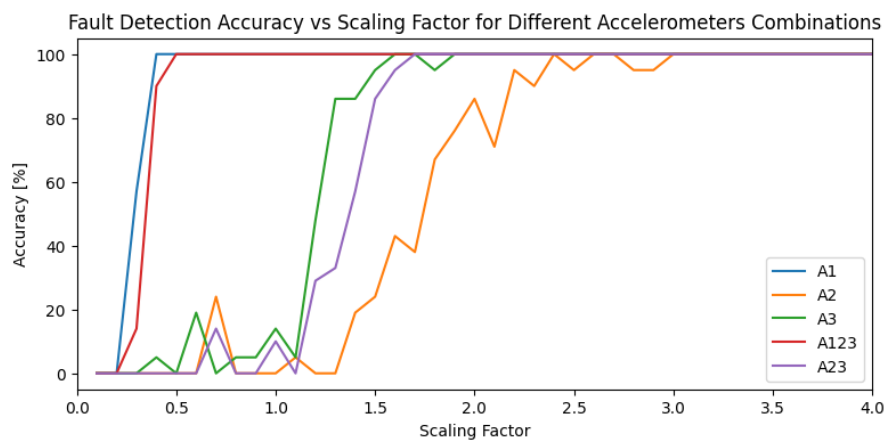


Figure 23: Fault detection accuracy vs scaling factor for different accelerometer combinations

5 Experimental Investigation

The purpose of the experimental testing is to gather real-world multi-sensor vibration data to evaluate the performance and robustness of the proposed fault detection model. While the simulated signals are useful for developing the detection model, experimental measurements are important for understanding how the detection model performs in the presence of machine noise, structural transmission paths, and unpredictable impulses.

The experimental test bench is designed to help answer three key questions related to the research objectives:

1. **Can the detection model reliably identify faults in real-world vibration data collected from an operating system?**

The test bench can provide both healthy and faulty conditions, allowing for the model's diagnostic capabilities to be assessed.

2. **Can the fault still be detected when the sensor is placed far from the fault source?**

Accelerometers are placed at increasing distances from the fault and along different structural paths. This enables evaluation of how fault signatures propagate across different parts of the test bench.

3. **Does using multiple accelerometers provide improved diagnostic information, or do the extra accelerometers only help with redundancy?**

By recoding the vibration signals from all accelerometers simultaneously, the setup enables comparison between single- and multi-sensor configurations.

The experimental vibration data include variations in the operating speed, fault severity, fault alignment, and externally introduced random impulses. These conditions aim to replicate realistic monitoring conditions in a controlled space.

5.1 Bearing Test Bench

All experimental vibration data used were collected from the bearing test bench, which is shown in Figure 24. The test bench is designed to provide controlled, consistent operating conditions while enabling multiple accelerometers to be placed at different points on the structure. The main components of the test bench are listed below:

- **Motor:** A DC electric motor is used to drive the shaft. The motor speed has been limited to 1000 RPM to prevent excessive wear and to ensure safe, repeatable operating conditions.
- **Coupling:** A flexible coupling is used to connect the motor to the shaft. The purpose of this is to transmit the torque from the motor to the shaft. While also compensating for some misalignment and absorbing high-frequency vibration that would otherwise unnecessarily load the motor.
- **Self-Aligning Bearing:** The self-aligning bearing supports the shaft close to the motor. This bearing is used to limit small angular misalignment to help keep the shaft aligned during operation. This will help with the repeatability of the vibration measurements.
- **Imbalance Disk:** This disk allows for small weights to be added to the shaft to introduce imbalance vibrations to the system. Imbalance testing falls outside the scope of this research, but the disk is part of the test bench and can be used in future studies.

- **Test Bearing Housing:** The bearing that is being tested is mounted inside the housing, which also allows for fast replacement of test bearings. This makes it possible to evaluate bearings with different fault severities under identical operating conditions.
- **Extender:** An adjustable extender is attached to the shaft to apply a constant and controlled radial load to the test bearing. By changing the length of the extender, the applied force can be varied.
- **Load Cell:** A load cell is attached to the extender to measure the magnitude of the radial force that is applied to the shaft. This ensures that all tests are performed at a known and repeatable loading condition.
- **Test Bench Base:** All components are mounted to the steel base. The base provides structural stability and enables consistent sensor placement across all tests.
- **Tachometer:** A tachometer is mounted near the shaft to measure its rotational speed. It measures one pulse per revolution, making it useful for estimating average speed.

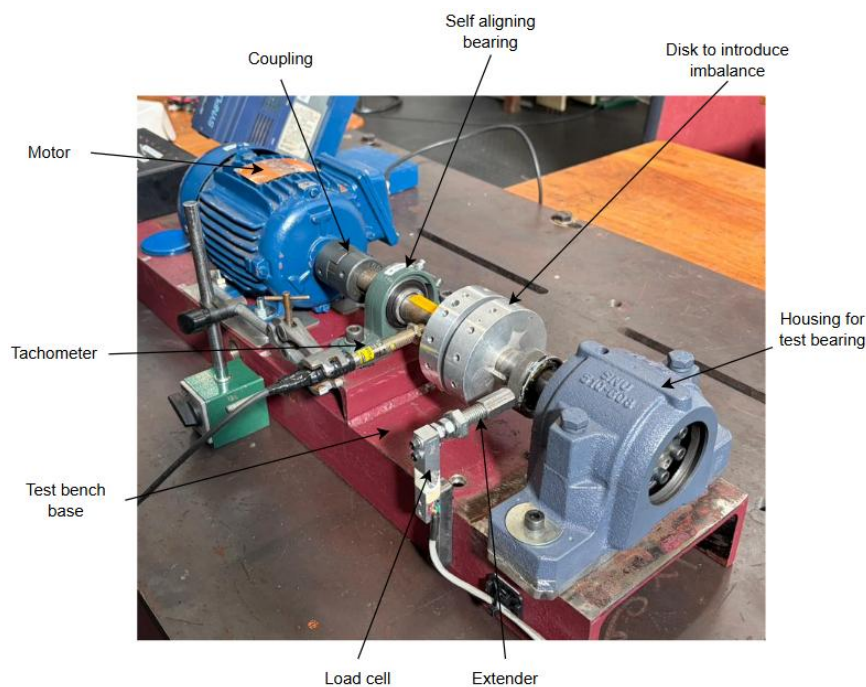


Figure 24: Labelled diagram of the controlled test bench

Five accelerometers were installed on the test bench for all conducted experiments. The sensors used are general-purpose industrial ceramic shear accelerometers with a sensitivity of 100 mV/g (model PCB IMI 603C01). For clarity, the accelerometers are labelled A1 to A5, allowing for the vibration signals from each accelerometer to be examined independently or in different combinations.

The positions of the accelerometers are shown on the left of Figure 25, with the right side showing the accelerometers attached to the test bench. Each accelerometer is attached to the test bench with a small stud secured with superglue. The accelerometer is then screwed to the stud to allow for repeatable mounting placements. This method of attachment provides a non-permanent, rigid connection between the sensor and the test bench, which is important for accurately capturing the high-frequency components of the signal.

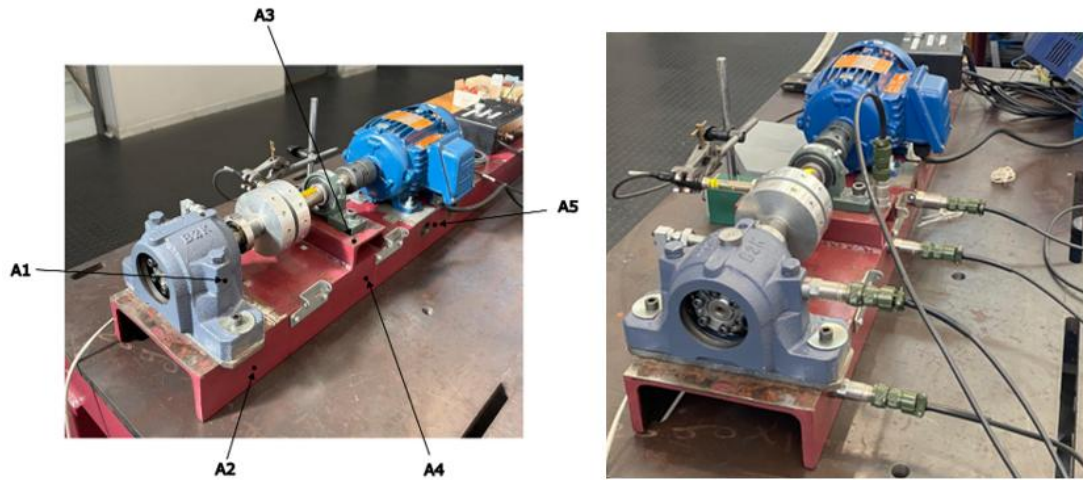


Figure 25: Accelerometer placement (Left), accelerometers on the test bench (Right)

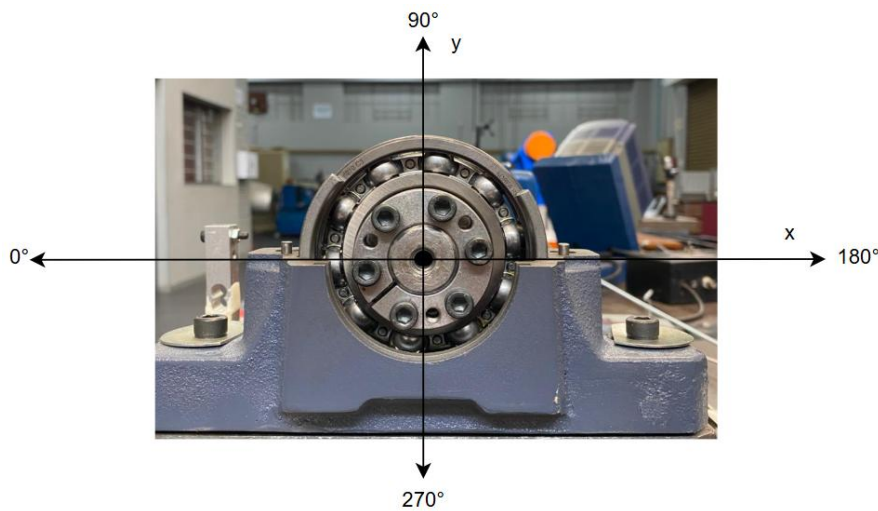


Figure 26: 6210 Deep-Groove Ball-Bearing in the test bearing housing

Figure 26 shows a side view of the 6210 deep-groove ball bearing used as the test bearing for all of the experimental measurements. A coordinate system has been assigned to the bearing to enable reference to specific points on the raceway during the analysis. This also helps to clarify the locations of the applied faults and the corresponding sensor placement and alignments. The applied load will act in the negative x-direction; the rolling elements will therefore carry the load at the 180° position on the bearing. This makes the 180° outer-race location the point of highest contact force and therefore the most likely position for an outer-race defect to generate strong impulsive vibration. Since accelerometers are highly directional, aligning them with the x-axis ensures they are most sensitive to the dominant fault-excitation direction.

Inner-race and rolling-element defects do not occur at a fixed spatial location, so their excitation is not tied to a single angular location. For this reason, their detection does not depend strongly on sensor orientation, whereas outer-race faults do, making x-axis alignment the most physically justified choice. The locations chosen are shown in Figure 25 with the justification for each placement given in Table 4.

Table 4: Justification for Accelerometer Placement

#	Location	Justification
A1	Directly on the faulty bearing housing	• Ideal fault measuring point with the highest expected SNR • Most likely to detect faults under all operating conditions • Used as the gold standard single-accelerometer reference
A2	On the test bench base, adjacent to the faulty bearing, aligned with the dominant fault axis	• Represents a situation where it is possible to mount the accelerometer close to the bearing • Same axis as the expected fault direction • Used as the realistic single-accelerometer reference
A3	On the base, as close as possible to the alignment bearing, measuring in the y-axis	• Captures fault energy that propagates through both the base and the shaft • Helps assess how well faults can be detected away from the source and off the primary axis
A4	On the base, aligned with the alignment bearing, measuring along the x-axis	• Captures fault energy that propagates through both the base and the shaft • Measures in-axis propagation of the fault along the base and shaft • Complements A3
A5	On the base next to the motor, aligned with the dominant fault axis	• Captures vibration where motor-induced excitation and bearing-fault excitation overlap, representing a worst-case environment • Acts as a far measuring point showing how fault signatures survive over distance and through multiple interfaces

An eight-channel Quantum X data acquisition system (model MX480B) is used to capture all experimental measurements. The system records the signals from all five of the accelerometers, the tachometer, and the load cell simultaneously at a sampling rate of 38 400 Hz. This ensures that detection results can be compared directly across different accelerometer combinations.

The tachometer and load cell measurements are recorded with the vibration data to verify that the operating conditions remain within the expected range during each test. The tachometer also provides reliable speed information that is required during the filtering and detection steps. The load cell is primarily used to confirm that the radial load applied during a test is consistent and repeatable.

5.2 Experiment Cases

To evaluate how well the detection model performs under realistic conditions, it is important to clearly define the operating range. This is to ensure that the experimental data provides meaningful information to help answer the research questions. The accelerometer placement from the previous section forms part of this. The different sensor locations allow the study of how the vibration signal propagates through different transmission paths and how sensor orientation and distance from the fault influence detection performance.

Figure 27 shows the bearing used to represent the different fault severities. All the bearings are the same 6210 deep-groove ball bearings to minimise variation between test cases. To ensure consistency between the bearings, the factory grease was removed and replaced with a small amount of oil. This is sufficient for the short operating periods used in the experiments. Removing the grease also prevents the metal shavings from the drilling process from becoming trapped inside the bearing and introducing unwanted damage.

Four bearings were prepared for testing:

- A healthy bearing with no damage
- A small fault (2mm diameter hole through the outer race)
- A medium fault (4mm diameter hole through the outer race)
- A large fault (6mm diameter hole through the outer race)

Simple drilled holes were chosen to represent the different fault cases, as they provide a clear and repeatable way to distinguish between different fault severities. As the bearing races are made from hardened steel, drilling through the outer race is difficult using standard workshop tools. For this reason, the bearings were sent to Xytho Engineering [42], who used their CNC machine to accurately drill the holes without causing any unintended damage.

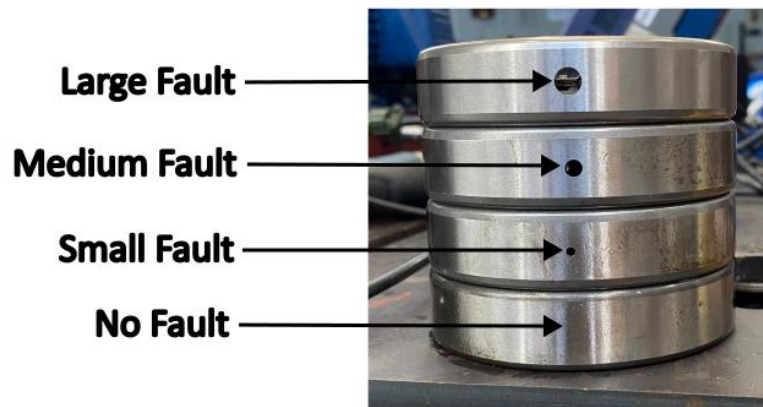


Figure 27: Different fault cases used

The effect of the fault orientation is also investigated by placing the fault at two different radial positions: 180° and 270° . This makes it possible to see how the alignment between the fault and the applied radial load influences the measured vibration signal. When the fault is positioned at 180° , it is directly in line with the radial load, and the rolling elements are expected to strike the defect with greater force. When the fault is positioned at 270° , it will not be aligned with the load axis, and the resulting impacts are expected to be weaker. The different testing positions aim to provide insight into how fault orientation will affect detectability across sensors.

To see how the vibration signal changes under different conditions, each fault case is tested at 500 RPM and 1000 RPM. These speeds provide a meaningful comparison: 1000 RPM is the test bench's maximum operating speed, while 500 RPM is a significantly lower speed that allows clear observation of how rotational speed affects the vibration response and the detection model's performance.

Each of the cases that are mentioned above is summarised in Table 5. Each of these cases will first be measured under normal operating conditions to establish a baseline for system behaviour. These baseline measurements will show how the detection model will perform when the test bench is running smoothly. To assess the robustness of the detection model, each case will also be tested in the presence of random impulses. The introduction of random impulses will show how well the model distinguishes fault-related impulses from unrelated external shocks.

This will result in 28 experimental cases used to help answer the research questions. A breakdown of all of the completed tests is shown in Appendix D

Table 5: Summary of experiment cases

Parameter	Values Tested	Notes
Applied Radial Load	18 kg	Load cell limited to 20 kg
Fault Severities	Healthy, Small, Medium, Large	–
Rotational Speeds	500 RPM, 1000 RPM	1000 RPM test bench limit
Fault Positions	180°, 270°	–

5.3 Experimental Procedure

This section outlines the procedure used to obtain each experimental measurement in this research project. The steps described here ensure that all tests were performed consistently and reliably, enabling meaningful comparisons across different fault cases, operating speeds, and impulsive conditions.

As mentioned earlier, all five accelerometers are used to capture vibration data for every test, together with the tachometer and load cell. All sensors record data at 38 400 Hz, ensuring sufficient information is captured and reducing the risk of aliasing in the measured signals. Each test is limited to a maximum duration of 25 seconds. This constraint is in place as the data is saved to an Excel spreadsheet, which has a limit of 1 048 576 rows; at a sampling rate of 38 400 Hz, this limit will be exceeded after approximately 27 seconds.

Each test followed the same sequence of steps to ensure consistency across all conducted tests. The procedure below describes the full process from starting up the test bench to saving the final data. The steps are as follows:

1. Set the motor speed

- The test begins by bringing the motor up to the required rotational speed.
 - Use the potentiometer on the control box to adjust the motor speed
 - Increase or decrease the speed until an appropriate target is met

2. Verify the rotational speed

- The speed of the shaft is measured to ensure it matches the intended test conditions
 - Measure the speed using a hand-held tachometer
 - Confirm the reading is within an acceptable range

3. Check the applied radial load

- The load cell reading is checked to ensure the correct radial force is applied
 - Verify that the load cells' output matches the required 18kg load
 - Adjust the length of the extender if necessary

4. Start the data acquisition

- Start recording with all sensors simultaneously using the CatmanEasy software
 - Start sampling at 38 400 Hz for all five accelerometers, the tachometer, and the load cell

5. Introduce impulsive events (only if required)

- Impulsive events are only added for the tests that include impulsive conditions
 - Skip this step for normal operating tests
 - For impulsive tests, strike the test bench with the ball of a ball-peen hammer at random times and intervals throughout the duration of the test

6. Record for the full test duration

- Each measurement is recorded for a maximum of 25 seconds

- Allow the system to run until the 25-second limit is reached and then end the data recording
- 7. Stop the motor**
 - The motor is switched off after each test to maintain consistency and protect the equipment
 - Bring the motor to a complete stop after termination of the data recording
- 8. Save and label the data**
 - The recorded data must be stored in the correct location with the correct case code described in Table 6Table 7
 - Save the file in the appropriate directory
 - Use the assigned code so that the test conditions can be easily determined

Each case has been assigned a unique code to keep track of all relevant information. These codes can be used to navigate the uploaded measurement files collected during the testing. The code includes 5 numbers, each representing a specific aspect of the test conditions. The meaning of each position in the code is summarised in Table 6.

Table 6: Meaning of each number in the for assigned name codes

Position of the number in the code	Meaning of numbers	Value meaning
1 st	Rotation Speed	1 – 1000 RPM 5 – 500 RPM
2 nd	Fault severity	0 – Healthy 2 – 2mm Fault 4 – 4mm Fault 6 – 6mm Fault
3 rd and 4 th	Fault Location	00 – Healthy 18 – 180° 27 – 270°
5 th	Presence of impulses	0 – No impulses 1 – Impulses

After all tests were completed, each test was passed through a vetting process. Any test was removed from the total samples if:

- The rotational speed was outside a $\pm 2.5\%$ margin of the intended speed, or
- The radial load was outside a $\pm 5\%$ margin of the intended 18kg load

This ensured that only valid and consistent samples were included in the final dataset, allowing for reliable comparisons across all operating conditions and fault cases.

5.4 Detection Pipeline

For applying the pipeline to the experimental data, the same procedure used for the numerical model is followed. However, this section focuses primarily on the results rather than the pipeline itself, as the methodology has already been demonstrated.

5.4.1 Filtering

The GMM models trained on the experimental data include features recorded at 500 RPM and 1000 RPM. This enables evaluation of how well the model detects faults when trained under different operating conditions. Ideally, the model should be able to identify a fault regardless of whether the test signal is obtained at 500 RPM or 1000 RPM. However, because the training data

includes two speeds, the filter sample time must be adjusted to ensure performance under both operating conditions. From Figure 28, it is clear that the optimal sample duration is approximately 3 seconds, and this value is therefore used for filtering. As with the numerical model, only the first 3 seconds of each signal are used during the filtering stage.

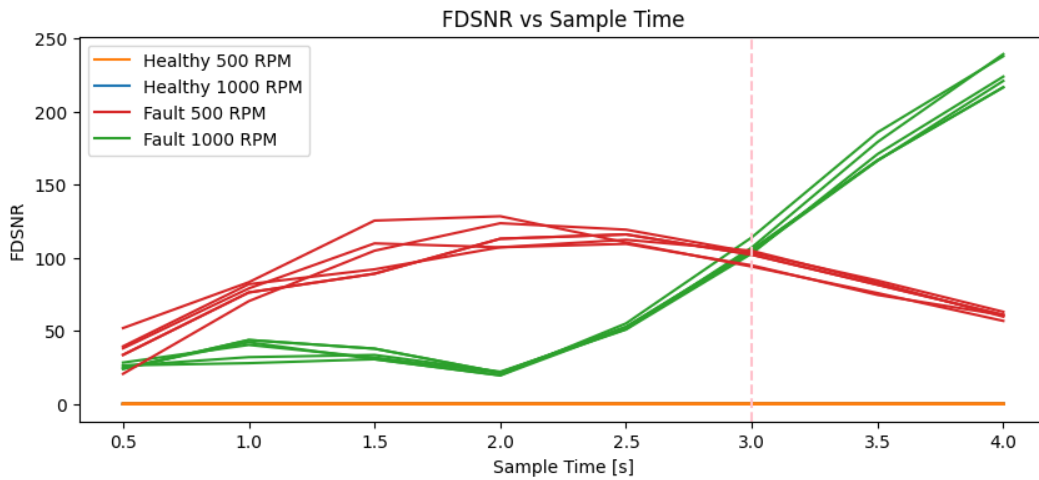


Figure 28: FDSNR vs Sample Time for 500 RPM and 1000 RPM vibration signals

5.4.2 GMM training

The healthy data used to train the GMMs follows an 80% split: 80% is used for training, and the remaining 20% is held out as unseen healthy test data. This allows the GMMs to be tested and validated on unseen healthy test data, ensuring they operate correctly. For each healthy operating condition, 20 25-second recordings were collected, and, as with the detection model, features are extracted from 1-second segments. As each recording was not exactly 25 seconds long, the rough number of training points is about:

$$0.8(20 \times 2)(25) \approx 800$$

Figure 29 shows the PCA plot for the single accelerometer A2 trained using both 500 RPM and 1000 RPM data. The PCA plot on the left reveals significant overlap between the weighted PCA features of the two speeds. This indicates that, under suitable conditions, the test bench's operating speed does not strongly influence the vibration features. This, in principle, means that a detection model may identify faults across different operating speeds without requiring training data for that specific speed. However, this behaviour is not guaranteed. As shown in Figure 30, where the accelerometer combination A3, A4, and A5 is used, the features from 500 RPM and 1000 RPM show little to no overlap. In this case, the two operating conditions are clearly distinct. This shows that it is possible for a model trained on a limited set of operating speeds to correctly detect anomalies at other operating speeds. This, however, depends strongly on sensor placement and the characteristics of the measured signal. It is also important to note that all training data was collected under steady-state conditions; this behaviour may not be valid for variable-speed operation.

Additionally, both PCA plots show that the model has fitted clusters to regions with minimal data. This shows how important it is to collect sufficient healthy data to ensure that the model can accurately generalise normal behaviour. If too little variation is captured during training, healthy points that fall outside the learned distribution may be incorrectly flagged as faults. Therefore,

gathering a wide and diverse range of healthy samples is important to ensure that new healthy data has the best chance of aligning closely with the learned healthy distribution.

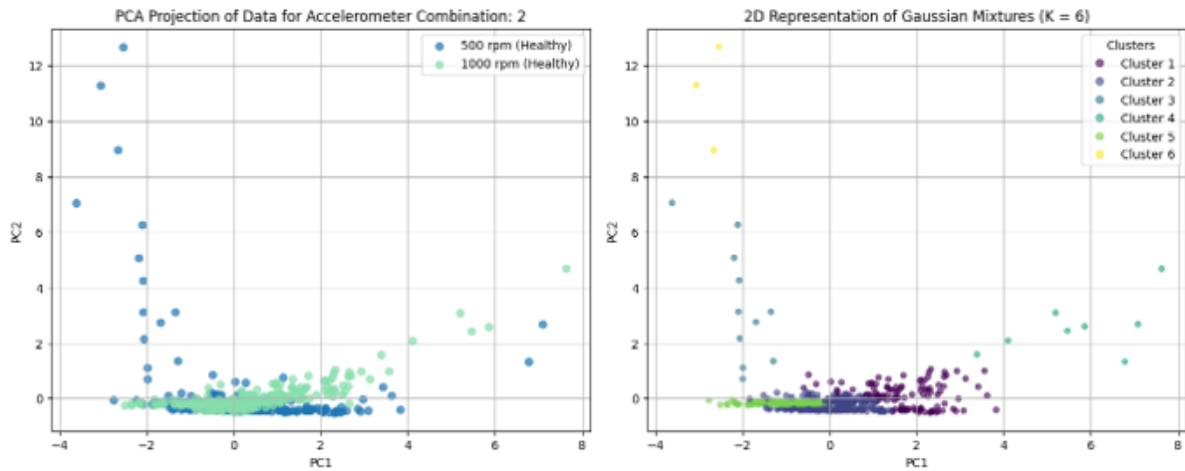


Figure 29: PCA plot for accelerometer 2 (left), 2D GMM plot for accelerometer 2 (right)

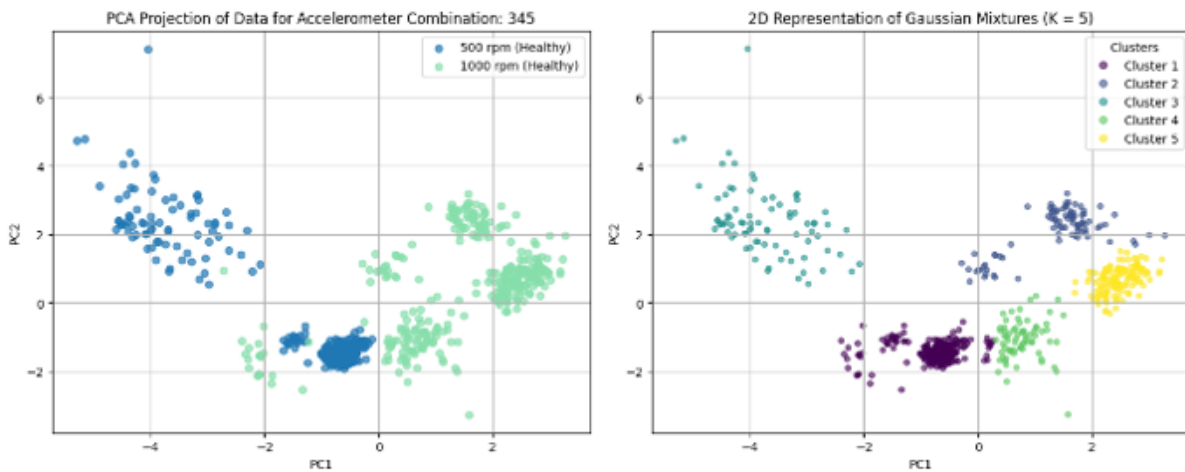


Figure 30: PCA plot for accelerometer combination 345 (left), 2D GMM plot for accelerometer combination 345 (right)

5.5 Detection Results

Figure 31–Figure 34 show the accuracy results for all experimental test cases for a range of different accelerometer combinations. Each accelerometer is first evaluated on its own to assess its performance and determine whether using multiple accelerometers provides any meaningful benefit for fault detection. The bar chart highlights broad trends in accuracy, while the heat maps provide a more detailed view of how different accelerometer positions and combinations perform under varying conditions.

For each accuracy score, three fault-case samples were randomly selected from the 10 fault cases operating under normal conditions to ensure consistency with the impulsive-fault tests, which have only three test samples. Since 80% of the healthy data is used for training, only four healthy samples remain for testing; therefore, using three samples per condition maintains consistency in the evaluation. This means that each accuracy value reflects the number of correct detections out of approximately 75 tested data points.

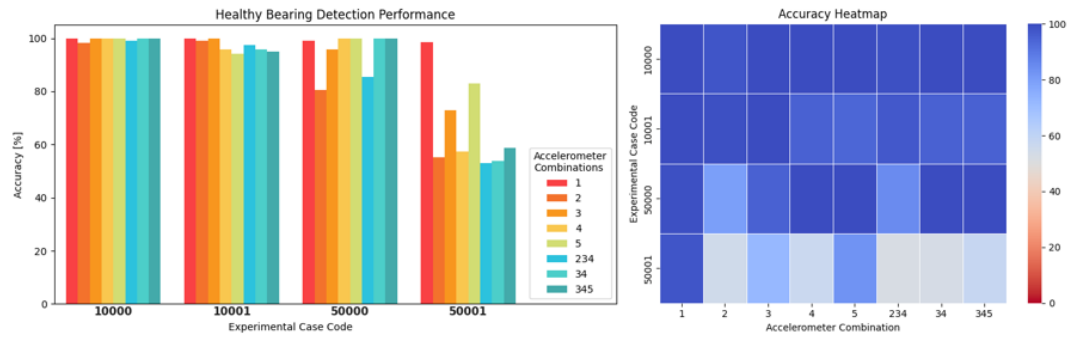


Figure 31: Healthy bearing detection performance for different accelerometer combinations

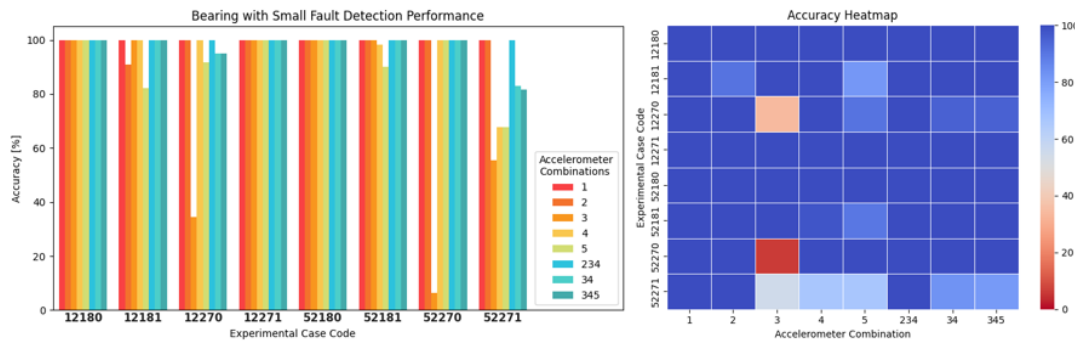


Figure 32: Bearing with small fault detection performance for different accelerometer combinations

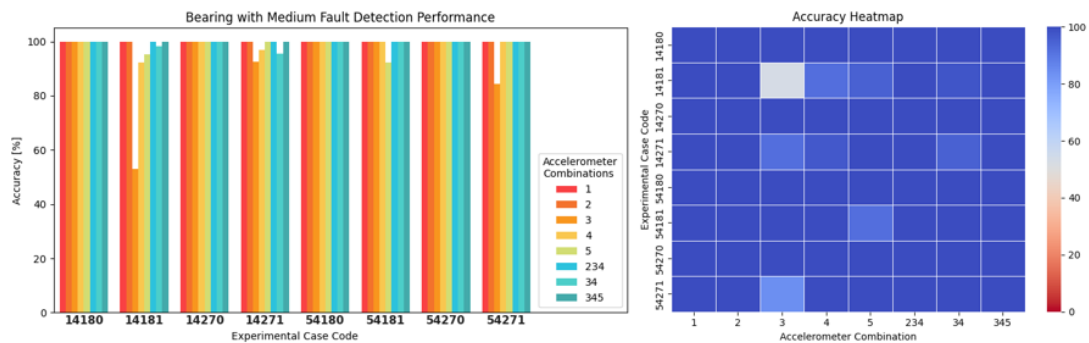


Figure 33: Bearing with medium fault detection performance for different accelerometer combinations

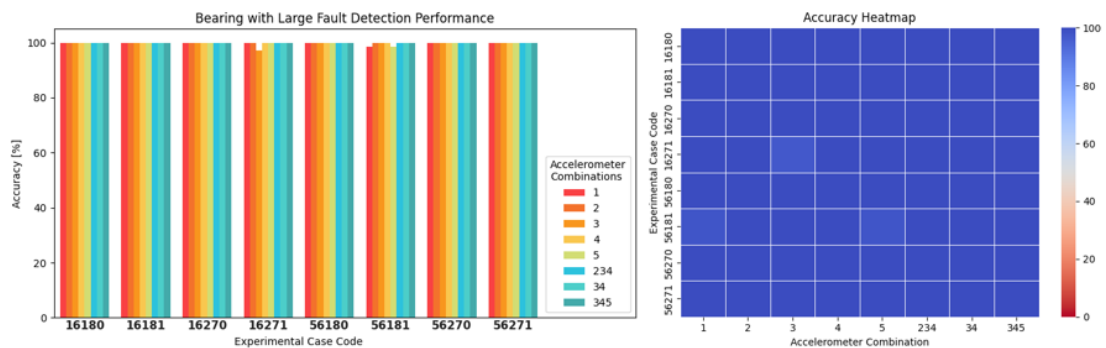


Figure 34: Bearing with large fault detection performance for different accelerometer combinations

When looking at Figure 31 it is clear that the detection model performed very poorly when the test bench was operating at 500 RPM with external impulses present in the signal. This shows that

something in the pipeline did not function as intended under these specific conditions. To narrow down the cause of this significant drop in accuracy, the first step is to assess how well the PES filtering removed or mitigated the impulsive peaks in the accelerometer signals.

Figure 35 show the raw and filtered signals for the healthy 1000RPM case using A2, which achieved nearly 100% accuracy. In contrast, Figure 36 shows the raw and filtered A2 signals for the 500 RPM case, where the model achieved only around 50% accuracy. Comparing the filtered signals for these two cases makes the source of the problem very clear. The PES filtering process did little to remove the effects of the random impulses in the 500RPM signal. The close-up view of the filtered signals shows this more clearly: the 1000 RPM signal is clear and well filtered, while the 500 RPM filtered signal still contains a large impulsive spike that has not been removed.

This shows that the PES filtering was not effective for the 500 RPM healthy impulsive case, as all but A1 performed very poorly. One possible reason for the poor performance is that only the first 3 seconds of the signal are used for filtering; results could be improved if each section of the signal were used for filtering. However, this seems to be unlikely as all of the other cases perform well; it is just the specific conditions for the 500 RPM and impulsive cases where the model fails.

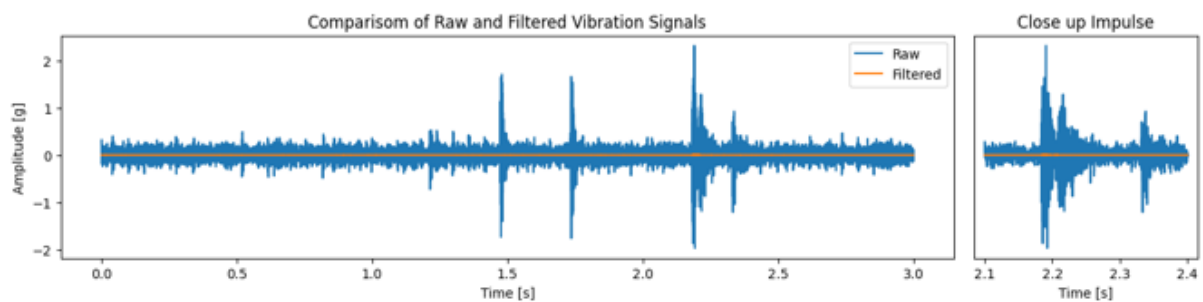


Figure 35: Raw and filtered signal for a healthy 1000 RPM vibration signal

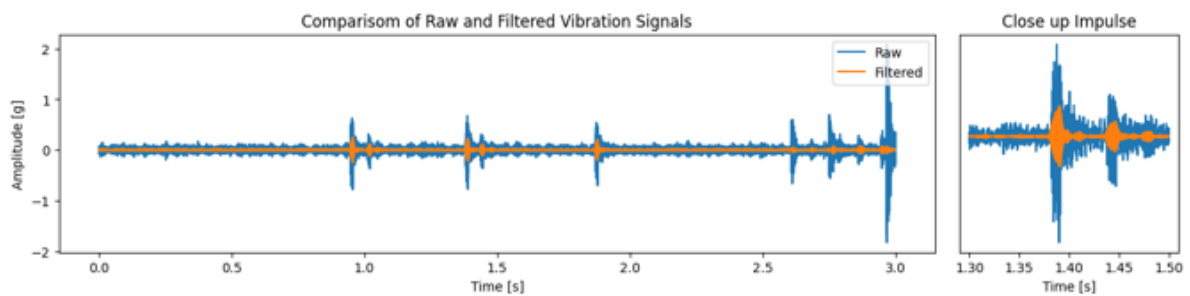


Figure 36: Raw and filtered signal for a healthy 500 RPM vibration signal

When examining the remaining results, the detection model performs well across most conditions, regardless of fault position or operating speed. The most common point of failure occurs in the individual readings from accelerometer A3. This is the only accelerometer not aligned with the axis of the applied load, and its performance does not improve even when the fault is positioned at 270°, where it is on the same axis as the accelerometer. This makes it clear that an accelerometer can provide richer information when it is aligned with the direction of the applied load, rather than simply being closer to the fault location. The accelerometers placed far from the fault, such as A5, consistently outperform A3, even when A3 is physically closer to the fault.

When comparing multiple-accelerometer results, it is clear that sensor combinations consistently outperform any single accelerometer, especially under impulsive conditions. For

example, when the small fault is positioned at 270 degrees, and the shaft is rotating at 500 RPM under impulsive conditions (case 52271), the individual accelerometers A3, A4, and A5 each achieve only around 50-60% accuracy. However, the accelerometer combination A3 and A4 achieves an accuracy of around 80%, while the combination of all three accelerometers achieves a slightly lower but still close-to-80 % accuracy. When an additional well-aligned accelerometer (A2) is included in the combination, the sensor combination A2, A3, and A4 achieves accuracy close to 100%.

This improvement from using multiple accelerometers is consistent across several tests and operating conditions. In many cases where a single accelerometer fails due to poor alignment, noise, or weak fault signatures, the combined accelerometer set shows a meaningful increase in accuracy. For this reason, using multiple accelerometers provides more valuable diagnostic information than using multiple sensors purely for redundancy. A multi-sensor configuration yields more robust and reliable detection, and even if one accelerometer in the combination performs poorly, the overall system remains robust and continues to detect faults effectively.

6 Conclusion

This research aimed to determine whether multi-sensor vibration measurements can meaningfully improve the robustness of bearing fault detection under harsh, impulsive conditions. A complete detection pipeline was developed, starting with an 8-DOF lumped-parameter numerical bearing model, followed by PES-based automatic filtering, feature extraction, feature-level fusion using weighted PCA, and anomaly detection using a Gaussian Mixture Model trained only on healthy data. The method was evaluated using both simulated and experimentally measured multi-accelerometer vibration signals.

The numerical and experimental results show that multiple accelerometers provide meaningful improvements in diagnostic performance, especially when the vibration signal comes from a noisy or impulsive environment. Across all datasets, the multi-accelerometer combinations consistently outperformed individual sensors or performed comparably. This confirms that the benefit of using multiple accelerometers is not simply redundancy but also improved ability to capture more information from different transmission paths.

An important observation is that accelerometer orientation is more important than the proximity to the fault. Sensors that are aligned with the axis of the applied load consistently produced clear fault signatures, even when positioned further from the bearing. The sensors positioned closer to the fault but not aligned with the load axis performed poorly. This behaviour was seen in both the numerical and experimental investigations, reinforcing the importance of properly aligned sensors in practical condition-monitoring systems.

The selected features were the FDSNR, the first harmonic magnitude for the PES, the spectral negentropy and the $\frac{l_2}{l_1} norm$ and they performed well under normal conditions. When the filtering stage successfully removed the impulsive peaks, the GMM achieved high accuracy across fault severities and operating speeds. However, some critical information was revealed: if the PES-based filtering fails to remove impulsive events from a signal, the detection model will fail regardless of the feature set or accelerometers used. This was very clear in the 500 RPM healthy cases, where the PESOgram did not reliably choose a band that would filter out the impulsive events. This resulted in the extracted features resembling faulty behaviour.

These findings do present some opportunities for future improvements. Refining the PES/PESOgram filtering method to improve its reliability under low speed or highly impulsive conditions. Investigating alternative filtering methods that might be more robust to random impulsive events, or exploring additional or alternative features that might be less sensitive to impulsive peaks in the signal.

Overall, the results demonstrate that the proposed multi-sensor detection method is effective under normal and most impulsive conditions, and that multi-accelerometer configurations significantly improve robustness. While the filtering stage remains the biggest limitation under harsh, impulsive conditions, the framework developed in this research provides a solid foundation for future improvements in vibration-based fault detection.

References

- [1] "Microsoft Copilot: Your AI companion." Accessed: Apr. 06, 2026. [Online]. Available: <https://copilot.microsoft.com/login>
- [2] "Grammarly." Accessed: Apr. 06, 2026. [Online]. Available: https://app.grammarly.com/?gad_source=1&gad_campaignid=23018054352&gclid=Cj0KCQjwkMjOBhC5ARIsADIdb3d-UWfeitlEJkbu7dnKz5Kj_Z4KF625d5ClBKVsc3l51NCeB4nLEJoaAimrEALw_wcB
- [3] "AEMT | Manufacturing." Accessed: Mar. 16, 2026. [Online]. Available: <https://www.theaemt.com/membership/member-sectors/manufacturing.html>
- [4] "Top 5 Faults Detected in Mining & Aggregates Operations | Machinery Lubrication." Accessed: Feb. 26, 2026. [Online]. Available: <https://www.machinerylubrication.com/Read/33033/top-5-faults-detected-in-mining-aggregates-operations>
- [5] R. B. Randall and J. Antoni, "Rolling element bearing diagnostics-A tutorial," 2011, *Academic Press*. doi: 10.1016/j.ymssp.2010.07.017.
- [6] "Master Vibration Analysis Methods for Precision Diagnostics." Accessed: Feb. 26, 2026. [Online]. Available: <https://reliamag.com/articles/master-vibration-analysis-methods-for-precision-diagnostics/>
- [7] "Vibration Analysis in Mining: Why Going Beyond Traditional Methods Matters - Razor Labs." Accessed: Feb. 26, 2026. [Online]. Available: <https://www.razor-labs.com/vibration-analysis-in-mining-why-going-beyond-traditional-methods-matters/>
- [8] Z. Huo, M. Martinez-Garcia, Y. Zhang, and L. Shu, "A Multisensor Information Fusion Method for High-Reliability Fault Diagnosis of Rotating Machinery," *IEEE Trans. Instrum. Meas.*, vol. 71, 2022, doi: 10.1109/TIM.2021.3132051.
- [9] R. B. Randall, "VIBRATION-BASED CONDITION MONITORING," New South Wales, 2021.
- [10] A. Jablonski, "Hybrid Model of Rolling-Element Bearing Vibration Signal," *Energies (Basel)*, vol. 15, no. 13, Jul. 2022, doi: 10.3390/en15134819.
- [11] Y. Liu, Y. Cheng, S. Yang, and J. Wu, "Vibration signal models for fault diagnosis of rolling element bearing," *Measurement (Lond)*, vol. 264, Mar. 2026, doi: 10.1016/j.measurement.2026.120303.
- [12] N. Sawalhi and R. B. Randall, "Simulating gear and bearing interactions in the presence of faults. Part I. The combined gear bearing dynamic model and the simulation of localised bearing faults," *Mech. Syst. Signal Process.*, vol. 22, no. 8, pp. 1924–1951, Nov. 2008, doi: 10.1016/j.ymssp.2007.12.001.
- [13] H. Cao, L. Niu, S. Xi, and X. Chen, "Mechanical model development of rolling bearing-rotor systems: A review," Mar. 01, 2018, *Academic Press*. doi: 10.1016/j.ymssp.2017.09.023.
- [14] I. Bagri, K. Tahiry, A. Hraiba, A. Touil, and A. Mousrij, "Vibration Signal Analysis for Intelligent Rotating Machinery Diagnosis and Prognosis: A Comprehensive Systematic Literature Review," Dec. 01, 2024, *Multidisciplinary Digital Publishing Institute (MDPI)*. doi: 10.3390/vibration7040054.

- [15] J. Prakash Kumar, P. S. Chauhan, and P. Prakash Pandit, "Time domain vibration analysis techniques for condition monitoring of rolling element bearing: A review," in *Materials Today: Proceedings*, Elsevier Ltd, Jan. 2022, pp. 6336–6340. doi: 10.1016/j.matpr.2022.02.550.
- [16] E. Wescoat, L. Mears, J. Goodnough, and J. Sims, "Frequency energy analysis in detecting rolling bearing faults," Elsevier B.V., 2020, pp. 980–991. doi: 10.1016/j.promfg.2020.05.137.
- [17] C. Peeters, J. Antoni, and J. Helsen, "Blind filters based on envelope spectrum sparsity indicators for bearing and gear vibration-based condition monitoring," *Mech. Syst. Signal Process.*, vol. 138, Apr. 2020, doi: 10.1016/j.ymssp.2019.106556.
- [18] E. Sejdic, I. Djurovic, and J. Jiang, "Time–frequency feature representation using energy concentration: An overview of recent advances," *Digit. Signal Process.*, Feb. 2008.
- [19] J. Cai and Y. Xiao, "Time-frequency analysis method of bearing fault diagnosis based on the generalized S transformation," *Journal of Vibroengineering*, vol. 19, no. 6, pp. 4221–4230, Sep. 2017, doi: 10.21595/jve.2017.18244.
- [20] Y. Yu, YuDejie, and C. Junsheng, "A roller bearing fault diagnosis method based on EMD energy entropy and ANN," *J. Sound Vib.*, vol. 294, no. 1–2, pp. 269–277, Jun. 2006, doi: 10.1016/J.JSV.2005.11.002.
- [21] J. Antoni, "Cyclostationarity by examples," May 2009. doi: 10.1016/j.ymssp.2008.10.010.
- [22] J. Antoni, "Cyclic spectral analysis of rolling-element bearing signals: Facts and fictions," *J. Sound Vib.*, vol. 304, no. 3–5, pp. 497–529, Jul. 2007, doi: 10.1016/j.jsv.2007.02.029.
- [23] M. Buzzoni, J. Antoni, and G. D'Elia, "Blind deconvolution based on cyclostationarity maximization and its application to fault identification," *J. Sound Vib.*, vol. 432, pp. 569–601, Oct. 2018, doi: 10.1016/j.jsv.2018.06.055.
- [24] J. Antoni, "The spectral kurtosis: A useful tool for characterising non-stationary signals," *Mech. Syst. Signal Process.*, vol. 20, no. 2, pp. 282–307, Feb. 2006, doi: 10.1016/j.ymssp.2004.09.001.
- [25] C. Boncelet, "Image Noise Models," in *The Essential Guide to Image Processing*, Elsevier Inc., 2009, pp. 143–167. doi: 10.1016/B978-0-12-374457-9.00007-X.
- [26] C. Yang, H. Li, S. Cao, and K. Zhang, "Detection and cancellation method of impulse noise in gearbox non-stationary vibration signals—a case study of port ship unloader," *Meas. Sci. Technol.*, vol. 34, no. 12, Dec. 2023, doi: 10.1088/1361-6501/acf0dd.
- [27] F. Kibrete, D. Engida Woldemichael, and H. Shimels Gebremedhen, "Multi-Sensor data fusion in intelligent fault diagnosis of rotating machines: A comprehensive review," Jun. 15, 2024, *Elsevier B.V.* doi: 10.1016/j.measurement.2024.114658.
- [28] S. Buchaiah and P. Shakya, "Bearing fault diagnosis and prognosis using data fusion based feature extraction and feature selection," *Measurement (Lond.)*, vol. 188, Jan. 2022, doi: 10.1016/j.measurement.2021.110506.
- [29] A. E. Chaleshtori and A. Aghaie, "Data fusion techniques for fault diagnosis of industrial machines: a survey," Guilan, Nov. 2022.

- [30] M. Žvokelj, S. Zupan, and I. Prebil, "Multivariate and multiscale monitoring of large-size low-speed bearings using Ensemble Empirical Mode Decomposition method combined with Principal Component Analysis," *Mech. Syst. Signal Process.*, vol. 24, no. 4, pp. 1049–1067, May 2010, doi: 10.1016/j.ymssp.2009.09.002.
- [31] F. Anowar, S. Sadaoui, and B. Selim, "Conceptual and empirical comparison of dimensionality reduction algorithms (PCA, KPCA, LDA, MDS, SVD, LLE, ISOMAP, LE, ICA, t-SNE)," May 01, 2021, *Elsevier Ireland Ltd.* doi: 10.1016/j.cosrev.2021.100378.
- [32] T. Adali, M. A. B. S. Akhonda, and V. D. Calhoun, "ICA and IVA for data fusion: An overview and a new approach based on disjoint subspaces," *IEEE Sens. Lett.*, vol. 3, no. 1, Jan. 2019, doi: 10.1109/LESENS.2018.2884775.
- [33] E. L. Russell, L. H. Chiang, and R. D. Braatz, "Fault detection in industrial processes using canonical variate analysis and dynamic principal component analysis," 2000. [Online]. Available: www.elsevier.com/locate/chemometrics
- [34] X. Jin, J. Fan, and T. W. S. Chow, "Fault Detection for Rolling-Element Bearings Using Multivariate Statistical Process Control Methods," *IEEE Trans. Instrum. Meas.*, vol. 68, no. 9, pp. 3128–36136, Sep. 2019, doi: 10.1109/TIM.2018.2872610.
- [35] S. Wang *et al.*, "Hybrid Deep Learning Framework for Anomaly Detection in Power Plant Systems," *Algorithms*, vol. 18, no. 11, Nov. 2025, doi: 10.3390/a18110704.
- [36] B. Chen *et al.*, "Product envelope spectrum optimization-gram: An enhanced envelope analysis for rolling bearing fault diagnosis," *Mech. Syst. Signal Process.*, vol. 193, Jun. 2023, doi: 10.1016/j.ymssp.2023.110270.
- [37] A. E. Chaleshtori and A. Aghaie, "A novel bearing fault diagnosis approach using the Gaussian mixture model and the weighted principal component analysis," *Reliab. Eng. Syst. Saf.*, vol. 242, Feb. 2024, doi: 10.1016/j.ress.2023.109720.
- [38] J. Antoni, "Fast computation of the kurtogram for the detection of transient faults," *Mech. Syst. Signal Process.*, vol. 21, no. 1, pp. 108–124, Jan. 2007, doi: 10.1016/j.ymssp.2005.12.002.
- [39] K. P. . Murphy, *Machine learning : a probabilistic perspective*. MIT Press, 2012.
- [40] "Gaussian Mixture Model Selection — scikit-learn 1.9.0 documentation." Accessed: Jun. 08, 2026. [Online]. Available: https://scikit-learn.org/stable/auto_examples/mixture/plot_gmm_selection.html
- [41] N. Sawalhi and R. B. Randall, "Simulating gear and bearing interactions in the presence of faults. Part I. The combined gear bearing dynamic model and the simulation of localised bearing faults," *Mech. Syst. Signal Process.*, vol. 22, no. 8, pp. 1924–1951, Nov. 2008, doi: 10.1016/j.ymssp.2007.12.001.
- [42] "Xytho Engineering – Since 2021." Accessed: Jun. 23, 2026. [Online]. Available: <https://www.xytho.co.za/>

Appendix A Scope of Work

Due to the limited time available to complete the research project, it will focus only on specific aspects of vibration-based fault detection under harsh and impulsive conditions. The aspects of vibration-based fault detection that will be included in the research project are:

- Development of a numerical bearing model that can simulate the vibration signals for healthy and faulty bearings.
- Simulation of outer race bearing faults and the vibration signals under changing operating conditions.
- Using Python to implement a fault detection method using:
 - Signal processing / numerical detection methods and/or
 - Using machine learning trained on extracted features.
- Investigation of harsh impulsive conditions such as:
 - Non-stationary speed variation,
 - Impulsive disturbances that are unrelated to damage.
- Compare the results obtained from:
 - A single accelerometer fault detection performance,
 - Multi-sensor fault detection performance (Sensors will include accelerometers and possibly an optical encoder).
- Validation of the fault detection method using real vibration data obtained from a controlled test rig.

Objectives

The main objective of this research is to investigate whether information from multiple accelerometers can be effectively combined to detect damage in rotating machines under harsh, impulsive operating conditions. The objectives to achieve this are:

Numerical Objectives

- Use numerical methods in Python to simulate multi-sensor vibration signals of healthy and damaged rotating machine bearings.
- Identify and implement suitable signal processing and/or machine learning methods that can use multi-sensor vibration data for damage detection.
- Simulate impulsive conditions with numerical methods and assess their impact on diagnostic performance.
- Use Python to successfully detect and identify bearing faults from the simulated vibration signal.

Experimental Objectives

- Experimentally gather multi-sensor vibration data from a rotating machine with healthy and damaged bearings.
- Simulate impulsive conditions and assess their impact on diagnostic performance.
- Use Python to successfully detect and identify bearing faults from the experimental vibration signal.

Comparison Objectives

- Compare single sensor versus multi-sensor fault detection performance with both numerical and experimental results.

Validate theoretical predictions against experimental results and identify the limitations of the multi-sensor approach relative to single-sensor methods.

Appendix B Gantt Chart

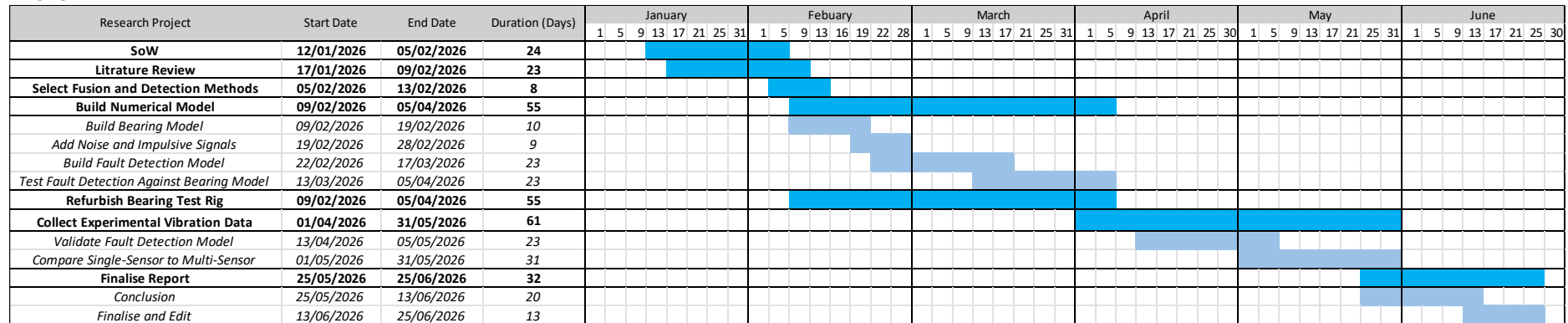


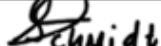
Figure 37: Gantt Char

Appendix C Meeting Log Card

**UNIVERSITY OF PRETORIA
DEPARTMENT OF MECHANICAL AND AERONAUTICAL ENGINEERING
RESEARCH PROJECT MRN 412/422**

MEETING LOG CARD

Student name and surname:	Ethan Jackson
Student number:	21481629
Study leader:	Dr. S. Schmidt
Year:	2026
Research project title:	Multi-Channel Signal Processing for Impulsive Fault Detection in Rotating Machinery

Date	Student signature	Study leader signature	Comments
08/12/2025			Initial discussion
03/03/2026			Scope of Work discussion
12/02/2026 - 23/06/2026			Weekly Meetings
	Stephan Schmidt, 2026/06/22		

ADDITIONAL COMMENTS:

Missed one weekly meeting on the 07/04/2026

Weekly meetings did not take place during test weeks

Appendix D : Extra Information and Link to all code and tests

All of the Python code and experimental data are included in the following link:

[MRN Files For Report](#)

Table 8 provides the full list of experimental cases used in this research. Each case has been assigned a unique code to keep track of all relevant information. These codes can be used to navigate the uploaded measurement files collected during the testing. The code includes 5 numbers, each representing a specific aspect of the test conditions. The meaning of each position in the code is summarised in Table 7.

Table 7: Meaning of each number in the for assigned name codes

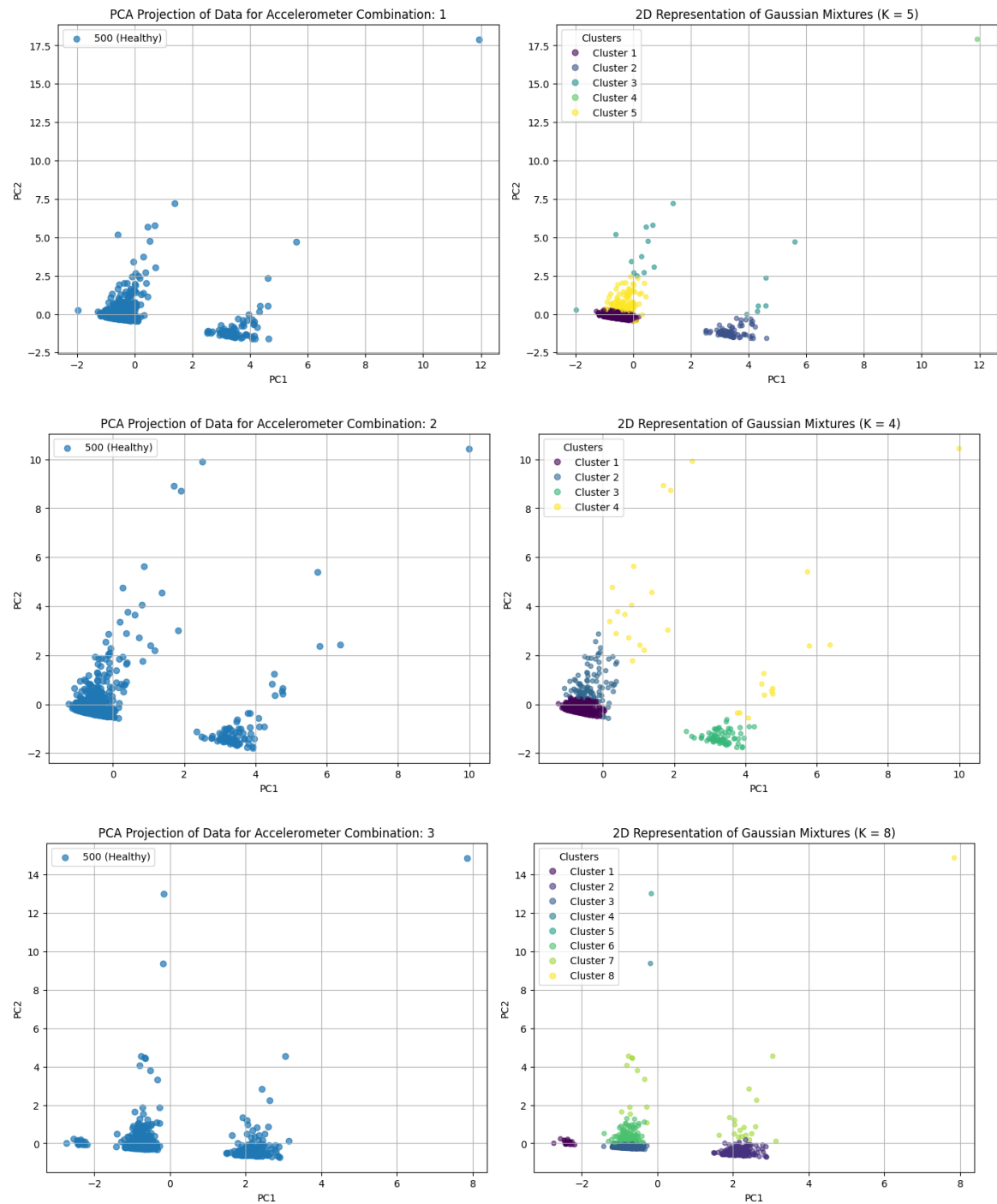
Position of the number in the code	Meaning of numbers	Value meaning
1 st	Rotation Speed	1 – 1000 RPM 5 – 500 RPM
2 nd	Fault severity	0 – Healthy 2 – 2mm Fault 4 – 4mm Fault 6 – 6mm Fault
3 rd and 4 th	Fault Location	00 – Healthy 18 – 180° 27 – 270°
5 th	Presence of impulses	0 – No impulses 1 – Impulses

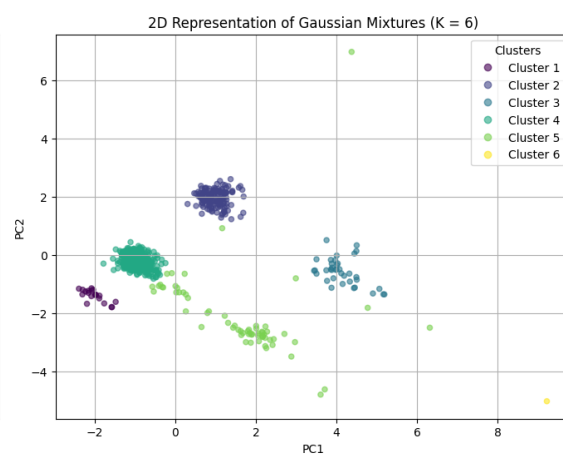
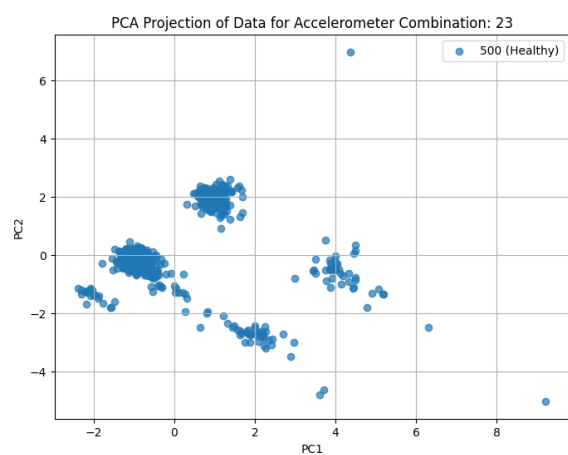
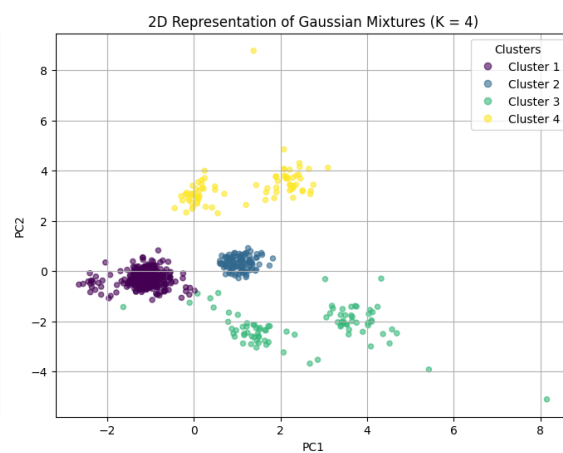
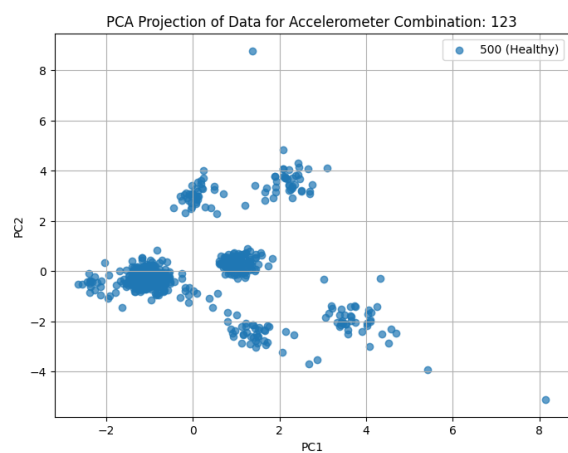
Table 8: Full list of experimental cases

Case Code	Fault Severity (mm)	Speed (RPM)	Fault position (°)	Impulses (yes/no)	Number of tests
50000	Healthy	500	Healthy	no	20
50001	Healthy	500	Healthy	yes	3
52180	2	500	180	no	10
52181	2	500	180	yes	3
52270	2	500	270	no	10
52271	2	500	270	yes	3
54180	4	500	180	no	10
54181	4	500	180	yes	3
54270	4	500	270	no	10
54271	4	500	270	yes	3
56180	6	500	180	no	10
56181	6	500	180	yes	3
56270	6	500	270	no	10
56271	6	500	270	yes	3
10000	Healthy	1000	Healthy	no	20
10001	Healthy	1000	Healthy	yes	3
12180	2	1000	180	no	10
12181	2	1000	180	yes	3
12270	2	1000	270	no	10
12271	2	1000	270	yes	3
14180	4	1000	180	no	10
14181	4	1000	180	yes	3
14270	4	1000	270	no	10
14271	4	1000	270	yes	3

16180	6	1000	180	no	10
16181	6	1000	180	yes	3
16270	6	1000	270	no	10
16271	6	1000	270	yes	3

Numerical PCA plots





Experimental PCA plots

